

ECS 205 A –Environmental Sciences

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic look.

Introduction to Environmental Sciences

Syllabus

ESC205A- Environmental Sciences

Credits: 2

Pre-requisites – No pre-requisites required for this course

Post Condition (on student capability after successfully completing the course):

At the end of the course, the student will be able to

- ☐ Understand the importance of environment and resource management
- ☐ Analyse and incorporate the essence of environment implications in their day today decision makin

- 1 Introduction to Environmental Science and Sustainable Development - Earth, Man and environment, ecosystem, pathways in ecosystem; Physic-Chemical and Biological factors in the environment. Concepts and principles of Sustainable Development 1.5 1 (project work based on field visit)*
- 2 Ecology and Biodiversity - Structure & functions of ecosystem, biodiversity and it's conservation 1.5
- 3 Natural Resources – Forest resources, water resources, energy resources, land resources and mineral resources 1.5
- 4-5 Air quality management – air pollution sources, air pollution & health impacts and air pollution control and management 3 Case study
- 5 Water quality management – water pollution, water and wastewater treatment and management 1.5

- 6 Solid and hazardous waste management – types of waste, health and environmental issues and waste management 1.5 Quiz
- 7-8 Environmental Management System - ISO 14000 series and ISO 14001, environmental policy, environmental performance evaluation, ecolabelling and corporate social responsibility 3 Case study
- 9 Energy and Environment – Renewable and non- renewable sources of energy and its impact on human health and environment 1.5 Field visit*

- 10 - 11 Environmental Decision Making Tools - Environmental impact assessment, Environmental risk assessment, Monitoring and evaluation technique, Life cycle assessment and Environmental modelling 3 Case study based learning
- 12 - 13 Contemporary Environmental Issues - Rain water harvesting, global warming and climatic change, climate vulnerability, carbon/ecological footprints, green economy and sustainable urbanism 1.5 Tutorial (assessment of carbon foot-prints - hands-on)

Field Visit *

- ▶ * A possible visit in NCR, Within the Campus or A virtual Visit through Video may be there at the end depending upon the time and other facilities.

Evaluation

- ▶ **Mid Term Examination 20 %**
- ▶ **Assignments – 15 %**
- ▶ **Quizzes – 5 %**
- ▶ **Project Work (Based on Field Visits) - 25 %**
- ▶ **Interactions in class and Attendance – 5 %**
- ▶ **End Semester Examination – 30 %**

Project Work

- Visit to a Local Area to Document Environmental Assets
River/Forest/Grassland*/Flora[#]/Fauna[#]/Hill*/
Mountain* etc.
- Visit to Some Local Polluted Site: Urban/Rural/
Industrial/Agricultural
- Study of Common Plants, Insects, Birds and *Basic
Principles of Identification
- Identification of Insects
- Identification of Common Insects
- Identification of Birds
- Identification of Some Common Plants
- Study of Simple Ecosystems – Pond, River,
Hill Slopes*, Delhi Ridge[#]

Texts/Other Resources

1. Gilbert M. M. and Wendell E. (2008). Introduction to environmental engineering and science, Prentice Hall, 2008.
- ▶ 2. Canter L.W. (1996) Environmental Impact Assessment, 2nd Edn. New York, McGraw Hill.
- ▶ 3. Odum E.P. (2007) Ecology: A Bridge between Science and Society.
- ▶ 4. Any book on Environment and Ecology by the Indian Authors or Foreign Authors, Internet, Environmental Reports, Research Papers or Current or Old News Items etc.

Introduction to Environmental Science & Sustainable development

As per Environment Protection Act 1986, Environment is defined as the sum total of water, air, and land and the inter-relationships that exist among them and with the human beings, other living organisms and materials

What is Environmental Science?

The study of how humans interact with
their environment

**Our environment is everything that
surrounds us, both natural and
man- made.**



Environment: the total of our surroundings

- **All the things around us with which we interact:**

- **Living things**

- Animals, plants, forests, fungi, viruses, etc.



- **Nonliving things**

- Oceans, clouds, soil, rocks

- **Our built environment**

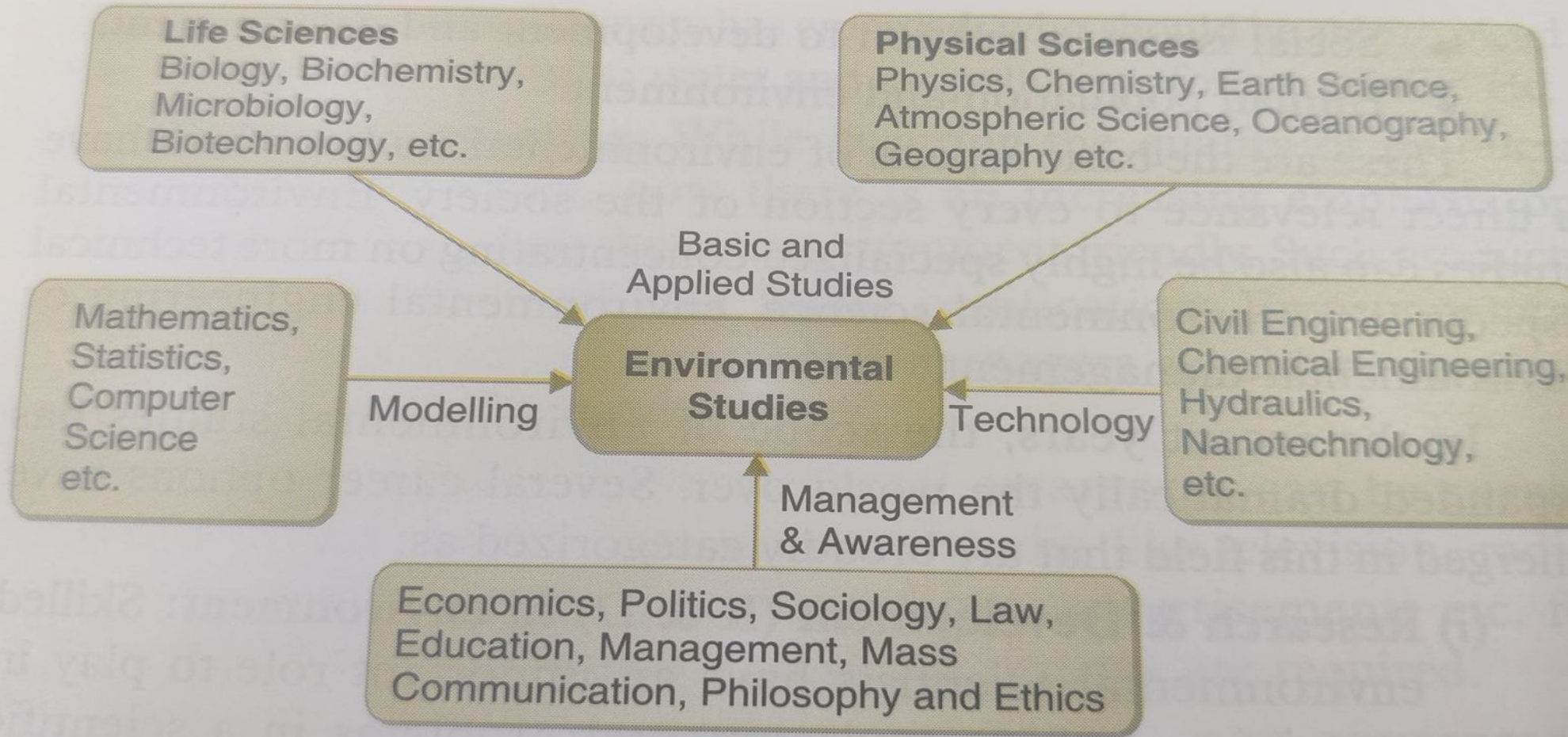
- Buildings, human-created living centers



- **Social relationships and**

SEGMENTS OF ENVIRONMENT

- **Atmosphere** – it is composed of several distinct layers such as troposphere, stratosphere, mesosphere and ionosphere.
- **Hydrosphere** – Water – Ocean
- **Lithosphere** – Solid part of the Earth crust.
 - The crust includes minerals, soil and rock.
 - Largest volume of Earth's interior – Mantle.
- **Biosphere** – It is the part of the earth which includes Air, Land, Water, Rock within which life occurs.



Earth, Man & Environment

- ▶ Humans impact the physical environment in many ways: overpopulation, pollution, burning fossil fuels, and deforestation. Changes like these have triggered climate change, soil erosion, poor air quality, and undrinkable water.
- ▶ Earth is like a mother to the mankind who are totally dependent on it. Age of earth is estimated to be about 4.6 billion years.
- ▶ Our ancestors depended on earth resources for their survival in the world and they started giving due importance to the resources such as trees, plants, sun , water , air etc. Plants, trees, herbs, minerals were also exploited as ancient medicines by them.

Video on Extinction of Dodos

- ▶ <https://www.britannica.com/video/Dodo-extinct-bird-island-Indian-Ocean-human-induced-extinction/-245100>
- ▶ Invaded by Dutch and Portuguese sailors

➤ 0 Today this human dependence is continuing as Ayurveda or Unani system of medicines. Cows were exploited to get milk. In fact the life form on this earth is being created through the recycling of the remains from the dead bodies, decayed plants, minerals etc. We are not sure about the rebirth but recycling through DNAs (N and P bases), genes, chemicals etc. etc. possibly through remains in the earth and these materials is happening.

➤ **INHERITED GENES- MITOCHONDRIAL GENES -37**

- Man has slowly developed techniques to exploit the nature, in fact earlier he was scared of the vagaries of nature like lightening, rains, fires, hot waves, storms, oceans etc.
- Necessity is the mother of invention.

- ▶ Agricultural practices were started almost 10000 years ago. Human brains developed science and technology to devise more and more products and processes to harness natural resources
- ▶ Several inventions took place with time. Industrial revolution started in 1760 and it is considered to have lasted up to 1840. Digitalization is happening in the Industry 4.0/5th period. Agricultural revolution was replaced by the industrial revolution and this has entered the space and computer age. We did not have mobiles commercially earlier till 1983 - AI, Space Technology etc.
- ▶ **Generative AI, Agentic AI, Google, OpenAI, Anthropic, Kimi K3 etc.**

- ▶ 1. Emission of greenhouse gases such as CO₂, CH₄, NO_x, water vapours etc., besides CFC, responsible for ozone layer depletion.
- ▶ 2. Hazardous chemicals (xenobiotics {i.e.,foreign to animal life) or anthropogenic compounds which are recalcitrant to biological degradation in nature) from agriculture and industries.
- ▶ 3. Production of nuclear materials.
- ▶ 4. Nuclear wastes.
- ▶ 5. Biological warfare.
- ▶ 6. Drones etc.

- ▶ 1. Emission of greenhouse gases such as CO₂, CH₄, NO_x, water vapours etc., besides CFC, responsible for ozone layer depletion.
- ▶ 2. Hazardous chemicals (xenobiotics {i.e.,foreign to animal life) or anthropogenic compounds which are recalcitrant to biological degradation in nature) from agriculture and industries.
- ▶ 3. Production of nuclear materials.
- ▶ 4. Nuclear wastes.
- ▶ 5. Biological warfare.
- ▶ 6. Drones etc.

▶ **INFORMATION ONLY**

- ▶ Federation of Indian Chambers of Commerce and Industry (FICCI) is organizing its 6th FICCI Technical Conference on Industrial Emission Monitoring & Pollution Control Technologies, on 24th -25th August 2026 at FICCI, Federation House, New Delhi (Non-Residential).
- ▶ Registration fee is to be paid.

- ▶ Air Pollution remains one of the most critical environmental and public health challenges globally, particularly in a rapidly developing economies like India. The conference will bring together regulators, industries, technology providers, and researchers to discuss the latest AI/IoT-enabled monitoring systems, regulatory compliance, and cost-effective emission control strategies. The brochure for the same is attached for your reference.
- ▶ The objectives of the conference are to:
 - ▶ • Discuss key provisions of regulatory developments and compliance requirements
 - ▶ • Highlight the latest technologies and innovations, Share industry best practices and case studies
 - ▶ • Address operational and maintenance issues of APCDs and emission systems in Industries
 - ▶ • Facilitate dialogue among regulators, industry, technology providers, & experts to promote effective industrial pollution management strategies.

Environmental Policy Integrated Climate (EPIC)

- ▶ The Near Extinction of Indian Vultures Led to the Death of a Half Million People
- ▶ A new EPIC study shows that when tens of millions of vultures were unintentionally poisoned in India, it removed "nature's sanitation service" and allowed deadly bacteria and infections to proliferate.

Ecology

- ▶ These activities have grave impacts on the environment and these are challenging the survival of man on this terrestrial ball. These may have to be controlled to make the life sustainable on this earth.

Ecology

The term ecology was described by Earnst Haeckel in 1869. Ecology deals with the study of interaction of organisms in their natural space with their surroundings.

Caution - Number of Slides are more because of bigger Font Size Please Note. This may also help you in reducing your dependence on materials. These may help you in acquiring more knowledge.

- ▶ Example is the interaction of tiny insect in a forest with other insects, animals, plants, soil, water, air etc.
- ▶ Thus, ecology is a science which deals with the interaction and relationship of organisms or a group of organisms.

Ecosystem AND Ecological Engineering

- ▶ Tansley in 1935 defined ecosystem as a self-regulating group of biotic community of species interacting with each other and with their non-living environment exchanging energy and matter.
- ▶ This helps us to understand the connection between organisms and their abiotic environment.
- ▶ Ecology is then the study of ecosystems.
- ▶ Ecological engineering is defined as the design, restoration, or creation of ecosystems, with a strong emphasis on ecosystem self-design and self-organization and controlling the chaos or disturbed situations. Self regulations / control or interdependence, cooperative or antagonistic domains etc. etc.

- **Ecosystem** - living things in a given area, non-living chemical and physical factors of their environment, linked together through nutrient cycle and energy flow

Types of Ecosystems

Natural

Artificial/Man-made

Terrestrial
(Forest, Grass
land, Desert)

Aquatic

Marine Fresh water

Lotic -river, stream or spring.

Lentic -lake, pond or swamp.

What causes ecosystems to change?

Natural causes:

1



Drought

Disease

2



3



Fire

4



What causes ecosystems to change?

Changes caused by humans:

1



Water pollution

2



Air pollution

3



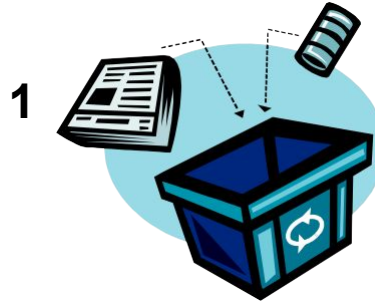
Land pollution

4



Construction

How can humans help to prevent changes in ecosystems?



Use resources wisely

Laws that control pollution



Clean up litter

Keep rivers and lakes



Ecosystems: Fundamental Characteristics

ABIOTIC components

- **Solar energy** provides practically all the energy for ecosystems.
- **Inorganic substances**, e.g., sulfur, boron etc. tend to cycle through ecosystems.
- **Organic compounds**, such as proteins, carbohydrates, lipids, and other complex molecules, form a link between biotic and abiotic components of the system.

Ecosystems: Fundamental Characteristics

BIOTIC components

- The biotic components of an ecosystem can be classified according to their **mode of energy acquisition**.
- **In this type of classification, there are:**
 - **Autotrophs and Heterotrophs**
- Organisms that produce their own food from an energy source, such as the sun, and inorganic compounds.
- Organisms that consume other organisms as a food source.

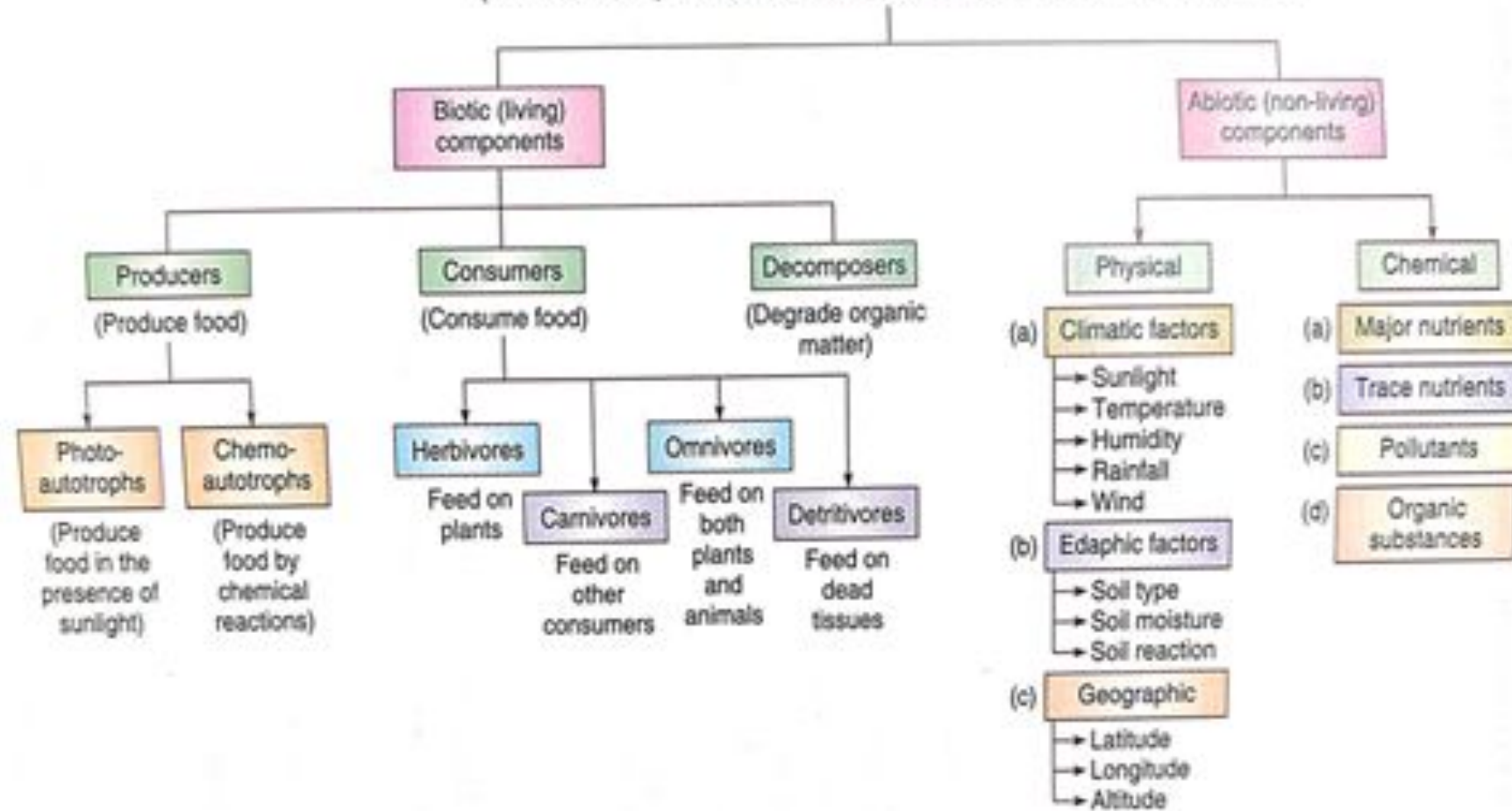
Structure of ecosystem

- ▶ The structure of ecosystem involves mainly two concepts i.e., food chains and food web through which energy and nutrients move through the ecosystems.
- ▶ Sun is the driving force of ecosystems. This structure is described in 5 levels :
- ▶ Producers
- ▶ Herbivores
- ▶ Carnivores
- ▶ Detrivores
- ▶ Decomposers

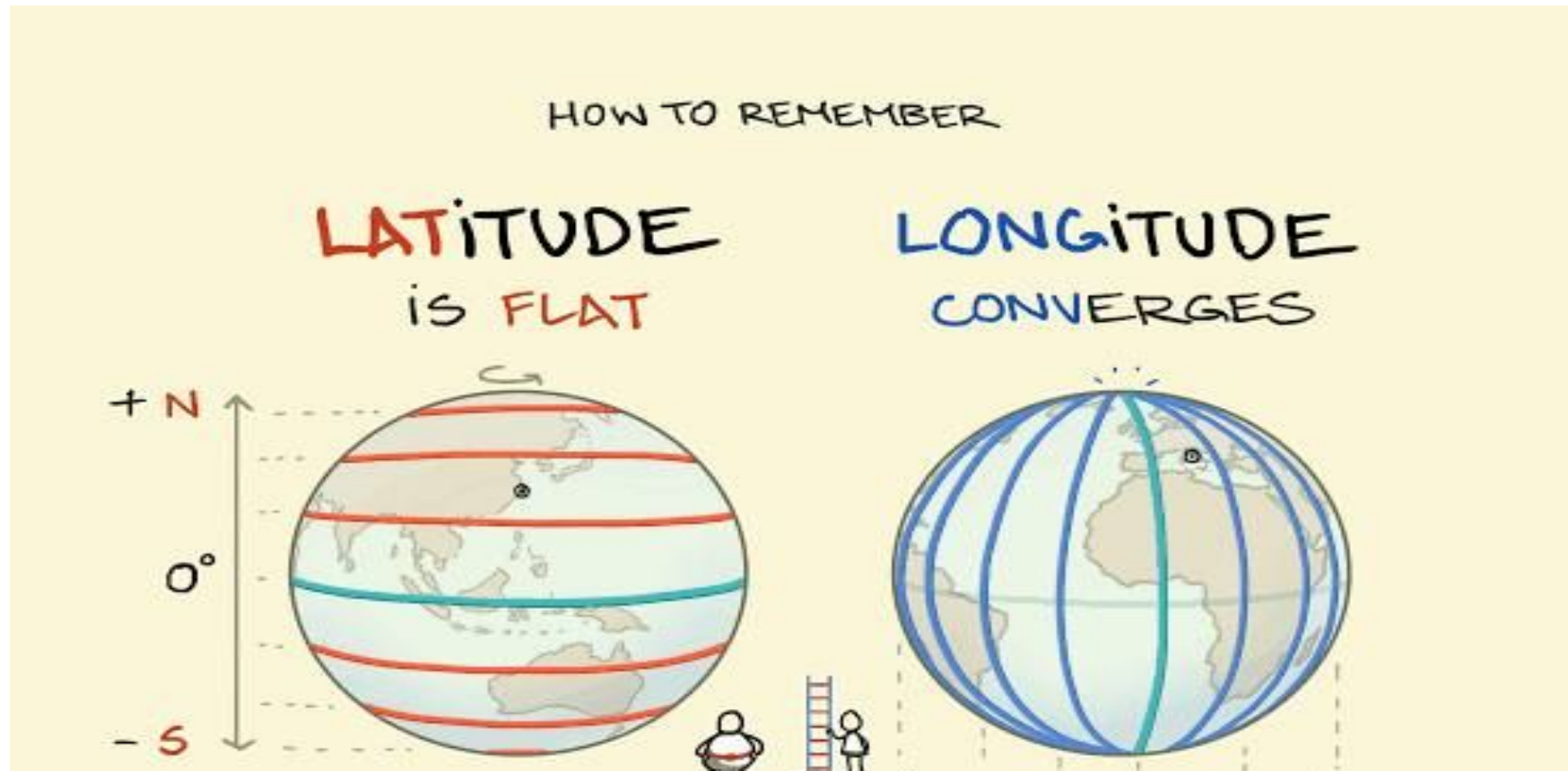
Ecosystems

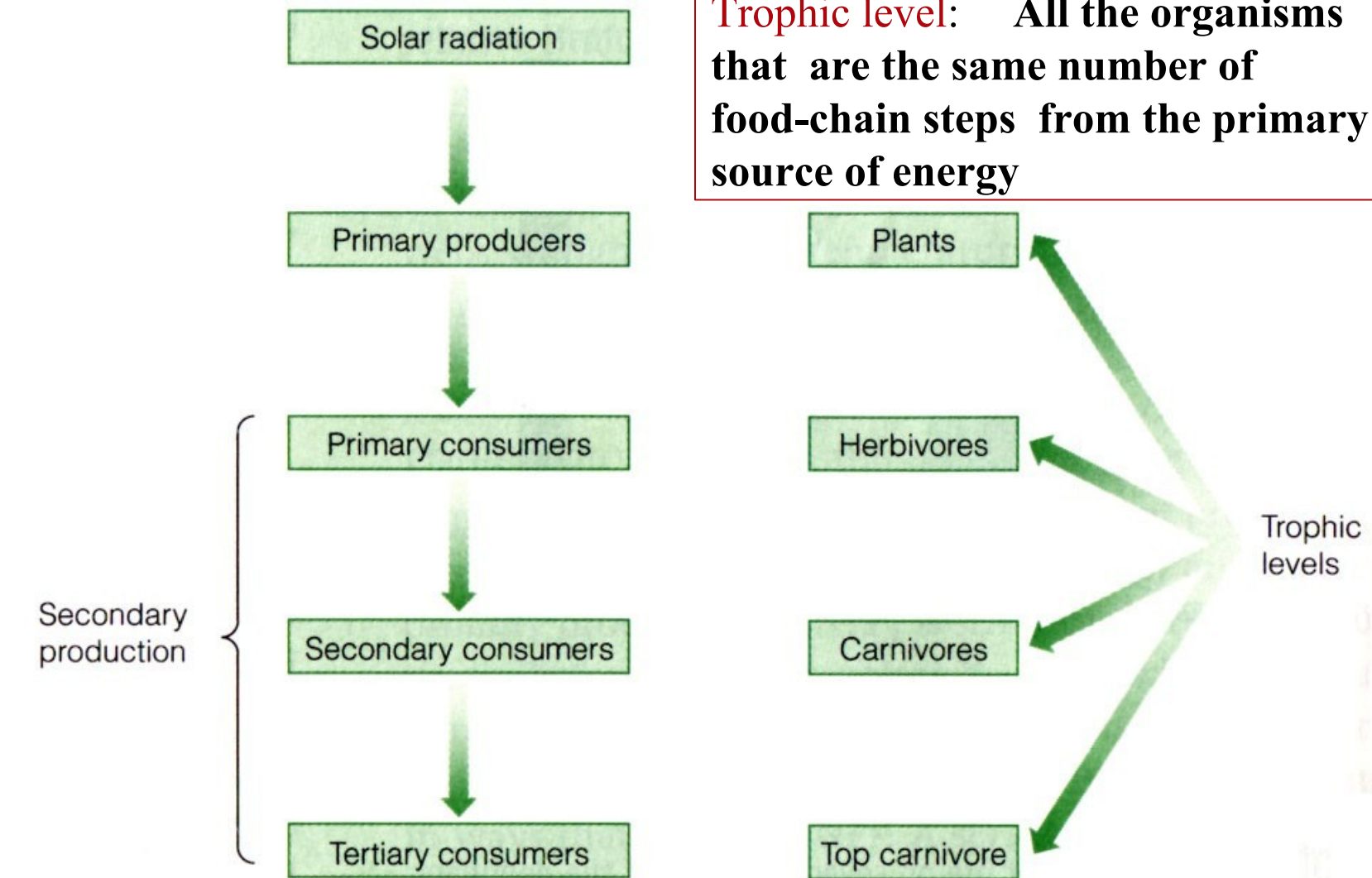
ECOSYSTEM STRUCTURE

(Communities of plants, animals and microbes interacting with the environment)



Altitude is the vertical height of an object or place above a given level, most commonly mean sea level. It tells you how high something is in the air, or how far up a place is located, and it is also used in math and science.





Trophic level: All the organisms that are the same number of food-chain steps from the primary source of energy

The schematic structure of a food chain. Each trophic level may contain many species.

Examples of interactions

Symbiosis is a term describing any relationship or interaction between two dissimilar organisms. The specific kind of symbiosis depends on whether either or both organisms benefit from the relationship.

For example, the close relationship between **Clown fish** and **sea anemones** :

The Stinging Tentacles of sea anemones - provide both partners a protection from predators and in return sea anemone is getting advantage as food from the Clown fish waste - Mutualism mostly.

Mutualism is defined as an interaction between individuals of different species that results in positive (beneficial) effects on per capita reproduction and/or survival of the interacting populations.. Example - Coral

Honey guide bird takes human to beehive & they eat wax left after the recovery of honey.
and algae, honeyguides bird guides humans to Beehives

Commensalism, in biology, a relationship between individuals of two species in which one species obtains food or other benefits from the other without either harming or benefiting the latter. Example - orchids growing on tree branches- get , Tree frogs use plants as protection. Golden jackals, trail a tiger to feed on the remains of its kills.

Antagonism, in ecology, an association between organisms in which one benefits at the expense of the other. As life has evolved, natural selection has favoured organisms that are able to efficiently extract energy and nutrients from their environment. Example opposing action of insulin and glucagon to blood sugar, Mosquitoes, pests and man.

Physico-chemical and biological factors in Environment

- ▶ Physico-chemical properties are the intrinsic physical and chemical characteristics of a substance. These include appearance, boiling point, density, volatility, water solubility and flammability etc
- ▶ **Physical factors of ecosystem and environment**

Critical physical factors are temperature, light, and hydrology (such as rainfall, soil moisture, flow rates, and sea level), as well as infrequent events that reshape ecological systems, such as fires, floods, and storms.

Cloudbursts occur when warm, moist air from low-lying areas is pushed upward along mountain slopes through orographic lifting. As this air rises to higher altitudes with lower atmospheric pressure, it expands and cools adiabatically.

Climate Modeling depends upon 7 factors

- ▶ Physical attributes reflect, in part, the influence of solar radiation. Solar radiation warms land and water masses, drives hydrologic cycles, and supports photosynthesis (which is essential to support biological systems).
- ▶ Biochemical factors include microbial, enzymatic and photobiological or electrobiological changes etc.

Factors that alter the critical chemical and physical characteristics of ecological systems include temperature, pH, electrochemical (redox) potential, and the transparency of air and water. pH of human system ?

Video on Environmental Engineering

- ▶ https://www.google.com/search?q=environmental+engineering+vedeo&rlz=1C1YTUH_enIN1045IN1045&biw=1507&bih=655&sca_esv=d827037c9c8be6ed&sxsrf=APpeQns_Bl8FZu9TIWsbMf7SLCm2HWWwDQ%3A1786537385301&ei=qWV8asP5EZGbhvcP4cGiwAc&ved=0ahUKEwiD8umxipuWaxWRjeEIHeGgCHgQ4dUDCBA&uact=5&oq=environme

Pathways of ecosystem :

- ▶ There are three major pathways through which plant or tree biomass is degraded into the forms that release nutrients again available to plants: **microbial decomposition; vertebrate herbivores; and wildfires.**
- ▶ **Here microbes, herbivores, and wildfires play a joint role in the functioning of ecosystems . Thus the diversity of mechanisms regulate the biosphere.**

Chemical factors of ecosystem

- ▶ Critical chemical attributes include levels of oxygen, nutrients, pH, salinity, and other chemicals in the environment.
- ▶ Biochemical factors include all biological attributes including microbial and other biological processes.

- ▶ Physical, chemical, and biological processes are **influenced by the amount and timing** of light including temperature and weather conditions, photoactivation of chemicals, mutations, and the timing of reproductive cycles. Solar radiation can have potentially harmful effects on some species.
- ▶ Reproduction and other activity patterns of species are often related to physical and chemical cues such as temperature, light, and salinity. They may also be linked to physical disturbances (such as periodic fires or floods).
- ▶ **Effect of temperature on fish hatching**

Food Chains

- ▶ Transfer of energy through eating and being eaten. Each step is called a trophic (relating to feeding and nutrition) level..
- ▶ The main source of energy for food chain is sun, dead animals or waste produced from living animals. On the basis of energy source there are two types of food chains :

Types of food chains :

- ▶ 1. Grazing food chain - Starting with plants etc
- ▶ 2. Detritus food chain - Starts with dead organic matter..

Trophic Levels

- A trophic (food and nutrition) level is the **position occupied** by an organism in a food chain.
- Trophic levels can be analyzed on **an energy pyramid**.
- **Producers** are found at the base of the pyramid and comprise the **first trophic level**.
- **Primary consumers** make up the **second trophic level**.
- **Secondary consumers** make up the **third trophic level**.

Finally, tertiary consumers make up the 4th trophic level.

Grasshopper camouflage - blend into fresh leaves for escape



- ▶ Living green plants produce food through photosynthesis with the help of UV light of sun by plants, trees, grasses. These are eaten by Herbivores. Carnivores depend upon herbivores for the same. This depends upon the sun and this autotrophic energy moves into herbivores and then into carnivores or omnivores. Most of the ecosystems in nature follow this food chain.
- ▶ Grass → Caterpillar or Grasshopper → Frog → Snake → Hawk

- ▶ Detritus Food chain
- ▶ This starts from the dead animals and microorganisms feeding on these (detritus) and their producers. This does not depend so much on solar energy and this depends on the organic matter produced in other system.



shutterstock.com - 1986232379

Leaf litter - Earthworms or insects - Sparrows or Crows or Black birds or Vultures - Hawk or Eagle

Detritus Food chain

- ▶ This starts from the dead animals and microorganisms feeding on these (detritus) and their producers. This does not depend so much on solar energy and this depends on the organic matter produced in other system.



- ▶ Leaf litter - Earthworms or insects - Sparrows or Crows or Black birds or Vultures - Hawk or Eagle

Saprophytes

- ▶ Saprotrophs are living things that get their food and energy by eating dead or decaying organic matter. They use external digestion by releasing special chemicals onto the dead matter to break it down, and then they absorb the simple nutrient.
- ▶ **Common Types of Saprotrophs :**
- ▶ **Fungi:** Mushrooms, bread molds like *Rhizopus*, and single-celled yeasts.
- ▶ **Bacteria:** Many soil-dwelling bacteria that break down organic waste.
- ▶ **Water molds:** Fungus-like organisms found in aquatic environment.

Food Web

- ▶ In fact feeding relationship in the ecosystem is complex. Some feed on producers and **some times carnivores also feed on dead organic material**. Some carnivores are dependent on varied diets, e.g., human feed on plants, animals, and fungi also. Such organisms are called **omnivores**, e.g., bears, birds, dogs, raccoons, foxes, certain insects, nightingales, and even humans.

- ▶ Most of the times the detritus food chain may inter-link with grazing food chain in a complex manner.
- ▶ Example is that the earthworm may feed on herbivores and on detritivore , waste produced from plants and animals and on dead bodies of plants or animals as a source of energy. The inter-linking food chain is like a mesh and it is known as Food web.
- ▶ This will be discussed in details later along with figure.

Functions of ecosystems

- ▶ 1. Energy flow
- ▶ 2. Nutrient cycling

Energy flow

- ▶ Autotrophic organisms use sun light to produce food and energy through photosynthesis. This energy is used by other trophic levels to remain alive and keep working.
- ▶ Nutrient cycling
- ▶ Nutrient cycling is a cyclic process that encompasses the movement of nutrients from the physical environment to living organisms and back to the environment- C,N,P, S cycling.

Chronobiology - Circadian Rhythms.

Body clocks-Master Clock: A small part of the brain called the suprachiasmatic nucleus (SCN) controls the rhythm.

- ▶ Circadian Rhythms (CRs)- are physical, mental, and behavioral changes that follow a 24-hour cycle. These natural processes respond primarily to light and dark and affect most living things, including animals, plants and microbes.
- ▶ It permits organisms to synchronize and assemble for daily or seasonal changes depending on the surrounding environmental conditions including dark and light.
- ▶ Circadian rhythms exist in species throughout the tree of life, from single-cell organisms (e.g., cyanobacteria, algae and Trypanosomes/ parasites) to humans.

CRs coordinate vital functions like sleep-wake cycles, hormone release, digestion, and body temperature, preparing your body for predictable changes in the environment and keeping your physical and mental health balance. Fish, animals, birds etc are affected **by light pollution and noise pollution..**

Pollution significantly affects circadian rhythms. Both light pollution and air pollution (such as ozone and PM 2.5 particulate matter) disrupt the internal biological clock, **alter the expression of circadian genes, suppress melatonin production, and increase the risk of metabolic, cardiovascular, and pulmonary disorders.** Leung and coworkers , Circadian Rhythms in Environmental Health Sciences: Current Environ

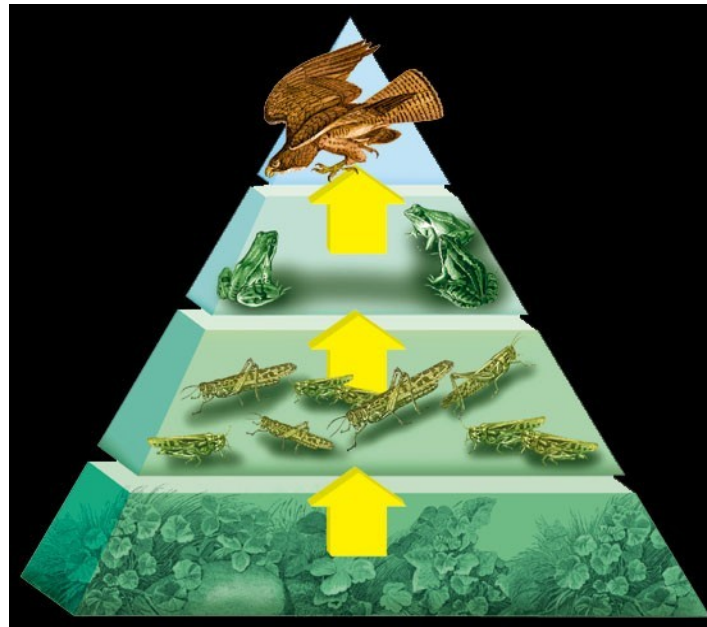


2. Nutrient cycling

- ▶ Bacteria and fungi decompose the dead bodies and waste produced from the living animals and even insects. The nutrients, chemicals and energy is circulated within ecosystem are known as **biogeochemical cycles**.
- ▶ C, N, S and P cycles etc.
- ▶ Recycling of humans
- ▶ Dynosaurs used the same water as we are drinking today.
- ▶ **Remains of dynosaurs?**
- ▶ **Harappa- 4,600 to 5,000-year-old human skeletons.**

Trophic Levels Found on an Energy Pyramid

- The greatest amount of energy is found at the base of the pyramid.
- The least amount of energy is found at top of the pyramid.



Source:
corpuschristisd.org/user_files/91702/Ecosystem.ppt

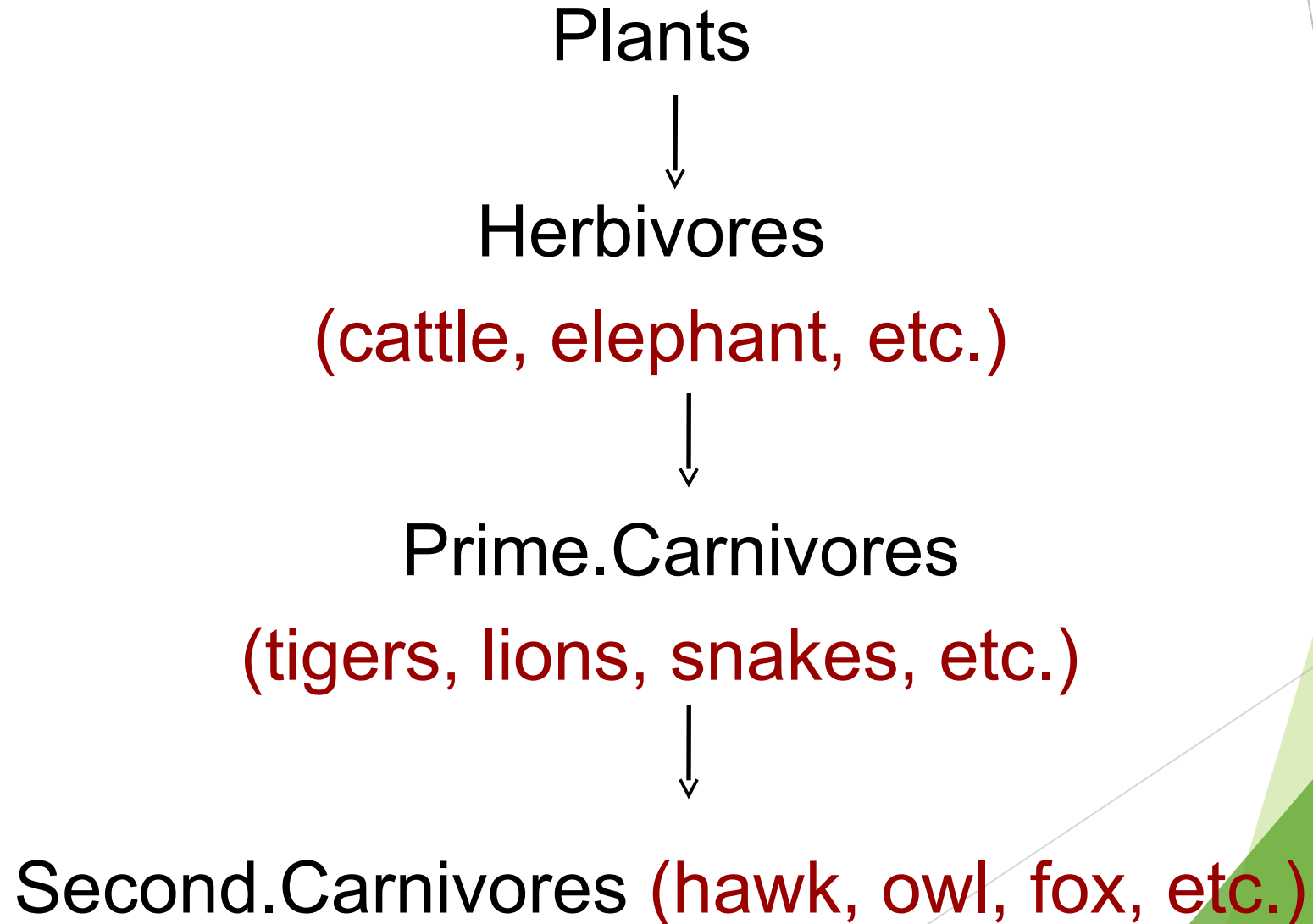
Food Chains

Transfer of energy and nutrients from one feeding group of organism to another

- The producers, consumers, and decomposers of each ecosystem make up a food chain.
- There are many food chains in an ecosystem.
- Food chains show where energy is transferred and not who eats who.

Types of Food Chains

1. Grazing food chain



Types of Food Chains

2. Detritus Food Chain - The organic waste and dead matter derived from grazing food chains are called detritus.

Dead Organic Materials (Decomposing organic matter)

Detrivores (Algae, Fungi, Bacteria, Earthworms, etc.,)

Chemical Energy -- Simpler Organic .

Compounds/ ATP- CO₂ + H₂O + Minerals + CaPO₄ etc.



Biomagnification-

- Biological Magnification- Hg eaten by fish & getting into human later on. DDT, the concentration of a substance (crosses) to increase at higher levels of the food chain.

– Ecological Balance :

- Maintaining and Regulating the population size
- Overpopulation - All organisms and even human
- Challenges of availability of Food and Pollution
- In addition, at each trophic level, the concentration of food chains keep on increasing,

Food Webs

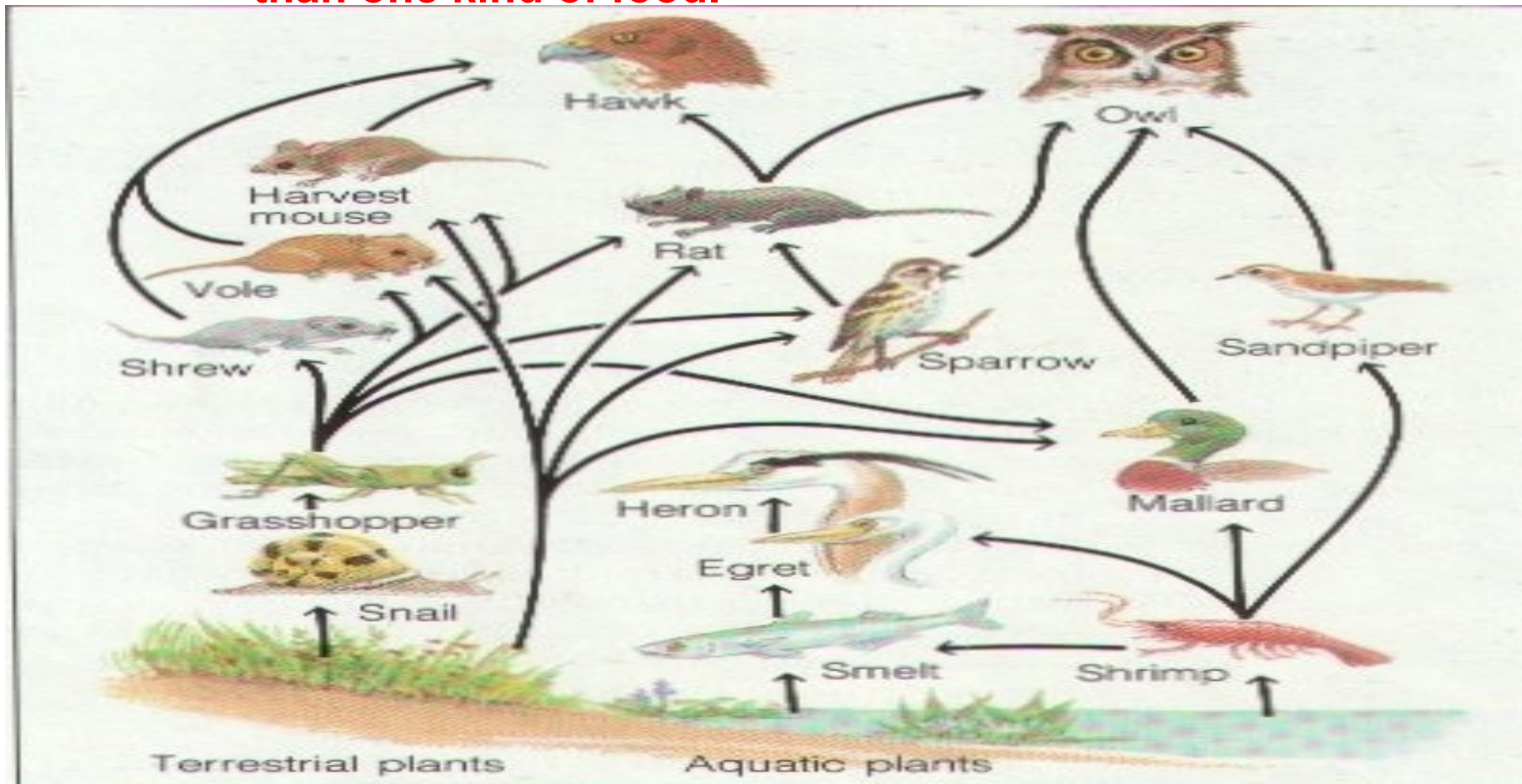
In ecosystems, some consumers feed on a single species, **but most consumers have multiple food sources.** Sometimes intensity of hunger and availability of food forces this tendency

Hawk/ Eagle eats both mouse and snake.

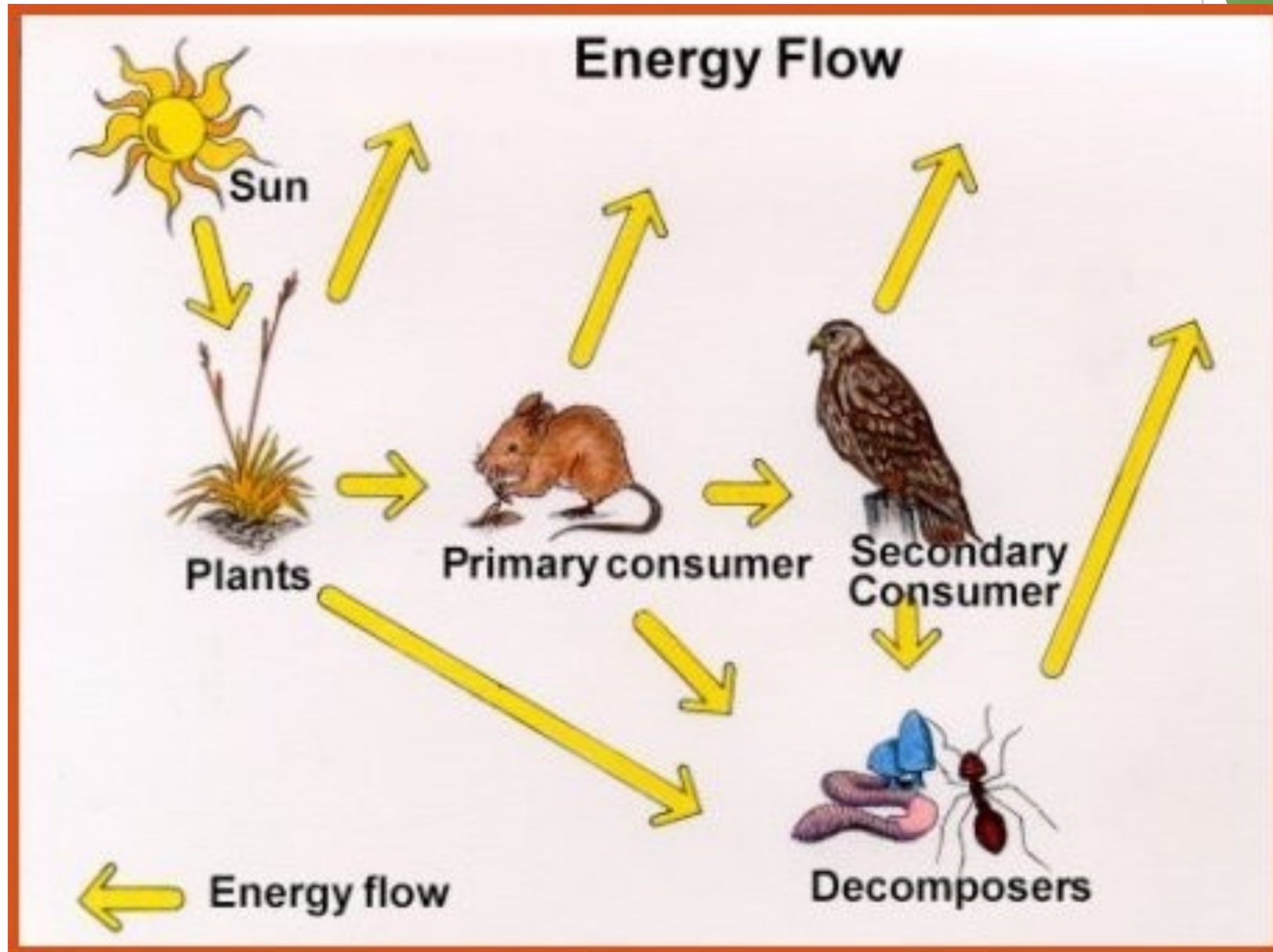
- In this way, individual food chain becomes inter- connected to form a food web.

Food Webs

- All the food chains in an area make up the food web of the area. Maintaining the stability of an ecosystem- Intense
- **Hunger driven Mad hunt for quick food. Something is better than nothing. Cross-grabbing .Most living things eat more than one kind of food.**



ENERGY FLOW IN ECOSYSTEM



ENERGY FLOW IN ECOSYSTEM

- All organisms must obtain a supply of energy and nutrients from their environment in order to survive.
- The transformations of energy in an ecosystem begin
- first with the input of energy from the sun.
- Because, it is the first step in the production of
- energy for living things, it is called “**Primary production**”.
 - Photosynthesis -- Chemical reaction where green plants use water and carbon dioxide to store the sun's energy in glucose.

ENERGY is stored in glucose. Glucose is stored as starch in plants and then metabolomics and biosynthetic reactions.

The energy contained within producers and consumers is ultimately passed to the decomposers that are responsible for the constant recycling of nutrients.

ENERGY FLOW IN ECOSYSTEM

• Thus, there is a one-way flow of energy through the biotic community and a cycling of nutrients between the biotic and abiotic components of the ecosystem

Energy flow cannot occur in reverse direction.

• The amount of energy decreases with successive trophic levels.

Only About 1% of energy from the sun is used by green plants & rest remains unutilized.

Similarly, there is loss of energy in each trophic level.

• The transfer of food energy between the organisms in an ecosystem can be tracked by constructing food chains, food webs, pyramids of numbers, biomass, etc.,

NUTRIENT CYCLES

- **Nutrient cycles involve storage and transfer of nutrients through different components of the ecosystem, so that the nutrients are repeatedly used.**
- **The cyclic movements of chemical elements of the biosphere between the organisms and environment are referred as “BIOGEOCHEMICAL CYCLES”**

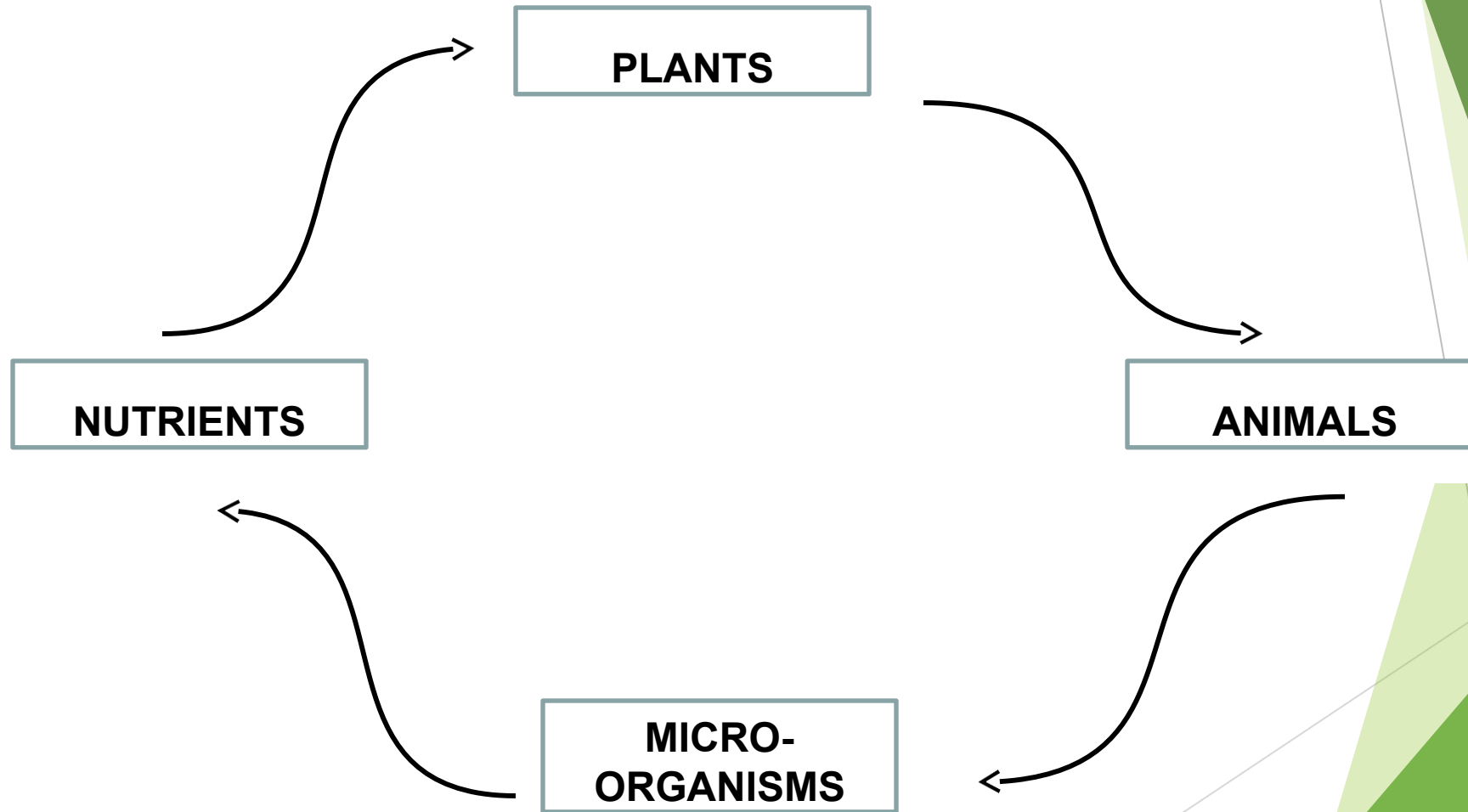
Human body is an ecosystem and not biome

- ▶ A **biome** is an area classified according to the species that live in that location. These form the niches for specific species, e.g., **rain forests, grassland, deserts etc.**

The **biosphere** is made up of the parts of Earth where life **exists**—all ecosystems. The biosphere extends from the deepest root systems of trees, to the dark environments of ocean trenches, to lush rain forests, high mountaintops, and transition zones like this one, where ocean and terrestrial ecosystems meet.

The Central Amazon Biosphere Reserve is a microcosm/representative of the Amazon's social, cultural, and ecological diversity. - Green Anaconda- 9 meters long and 230 kgs.

CYCLE OF NUTRIENTS



BIOGEOCHEMICAL CYCLES

- It involves biological, geological and chemical systems and all are interlinked through a cyclic chain.
- It is the complete pathway that a chemical element flows from the atmosphere, water, rock or soil to the living organisms and again back to the atmosphere, water, rock or soil.
- The return of chemical elements from living organisms to abiotic component is called Mineralization.

- There are 4 different biochemical cycles
 - Hydrological cycle (water)
 - Carbon cycle
 - Nitrogen cycle
 - Phosphorous cycle
 - Sulfur Cycle

HYDROLOGIC CYCLE

- In this cycle, fresh water evaporates and condenses on the earth. Oceans are the main source of evaporated water, which leaves behind salts.
- Water also evaporates from fresh water bodies, from land and plants.

HYDROLOGIC CYCLE

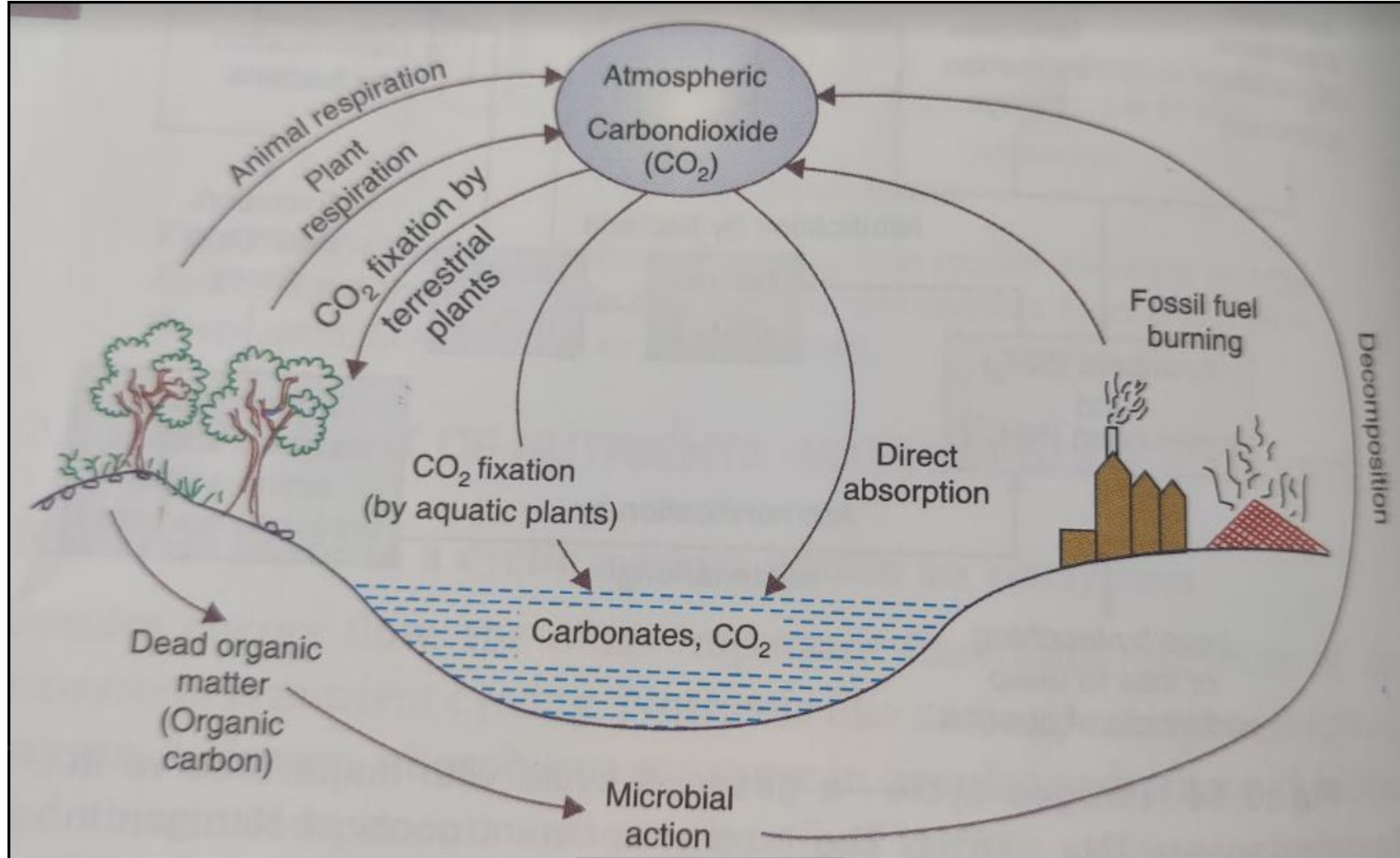
The steps involved in hydrologic cycles are

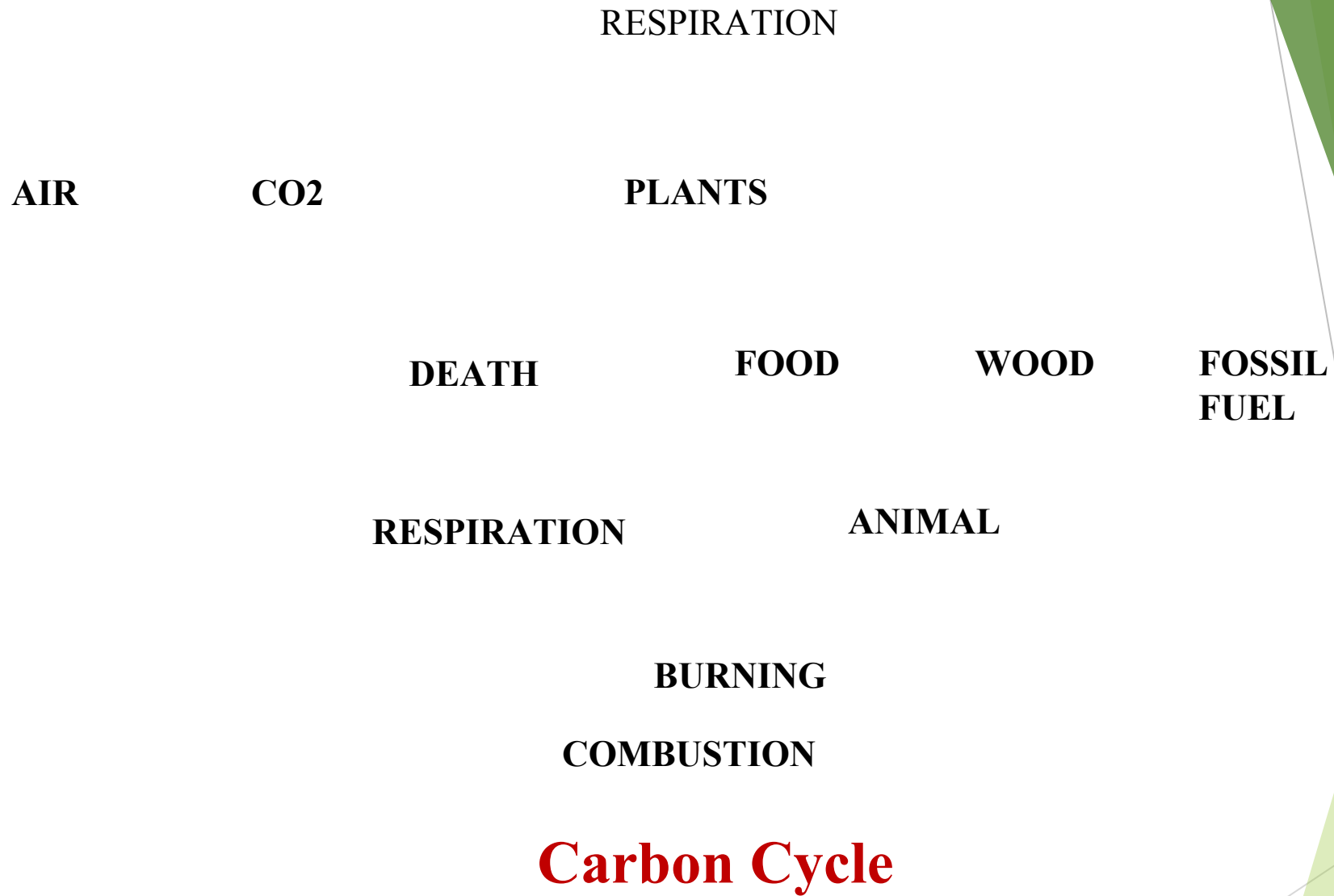
- evaporation**
- condensation**
- infiltration**
- runoff**
- Precipitation**
- Cloud bursts**

CARBON CYCLE

- Carbon enters plants as CO_2
 - Bacteria and enzymes process carbon in a fashion that allows it to be recycled.
 - Obtain energy from the molecules, and convert carbohydrates to carbon dioxide as a result of respiration.
- Photosynthesis removes carbon from the abiotic environment (fixes carbon into organic molecules)
- Carbon moves through food chain through consumption of one organisms by another
- Cellular respiration, combustion, and erosion of limestone return carbon to the atmosphere, water and abiotic environment.

Carbon cycle





The source of atmospheric carbon dioxide is variable but only plants can utilize atmospheric carbon directly

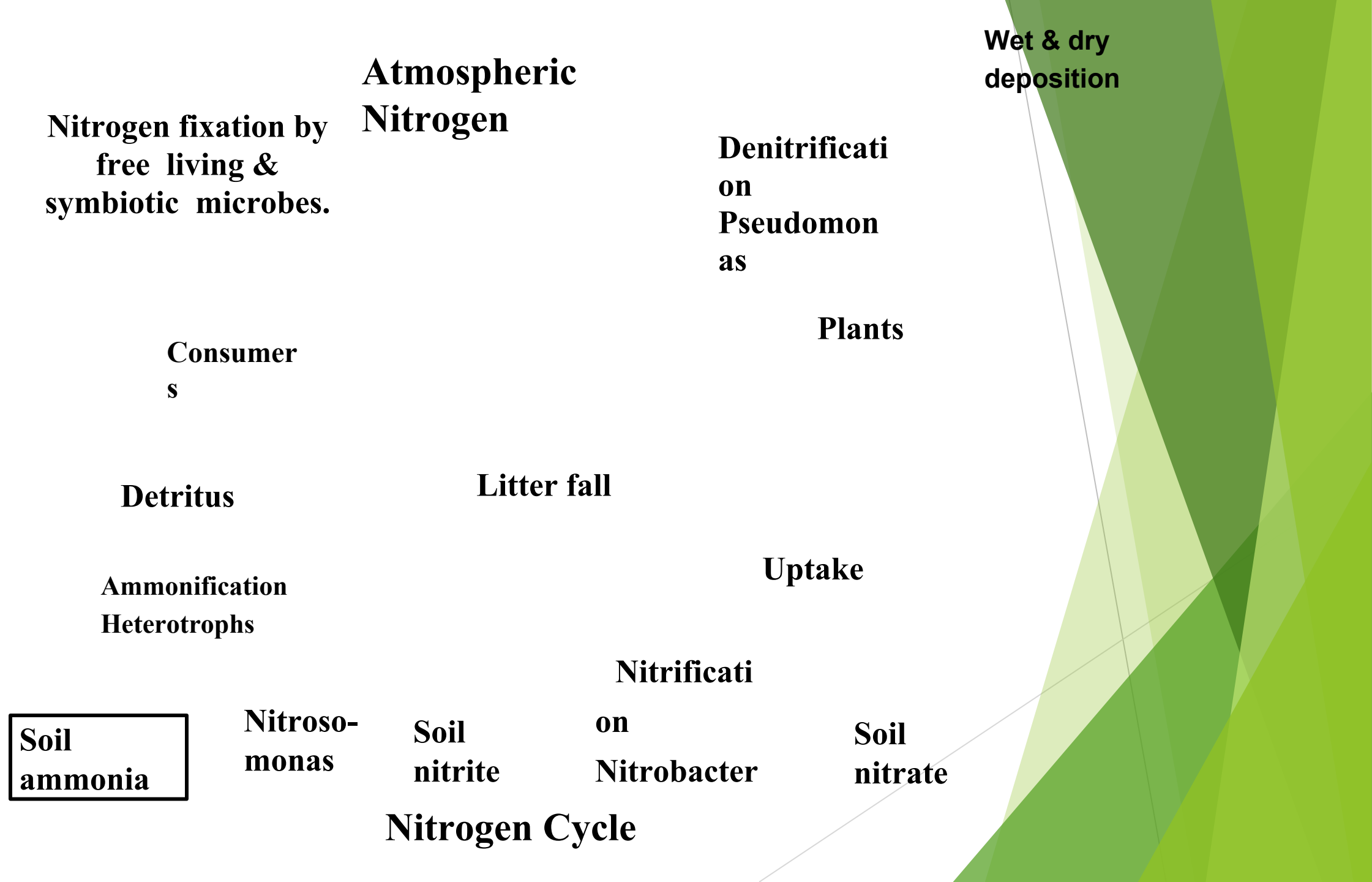
NITROGEN CYCLE

- **Nitrogen is crucial for all organisms**
 - Nucleic acids and DNA, RNA etc.
 - Proteins
 - Chlorophyll
- **Nitrogen- 78% in Atmosphere**
- **N₂ is very stable and must be broken apart by organisms, combined with other atoms into a usable form.**
- **The process of entering atm. Nitrogen into the organism and again back to the environment completes the Nitrogen**

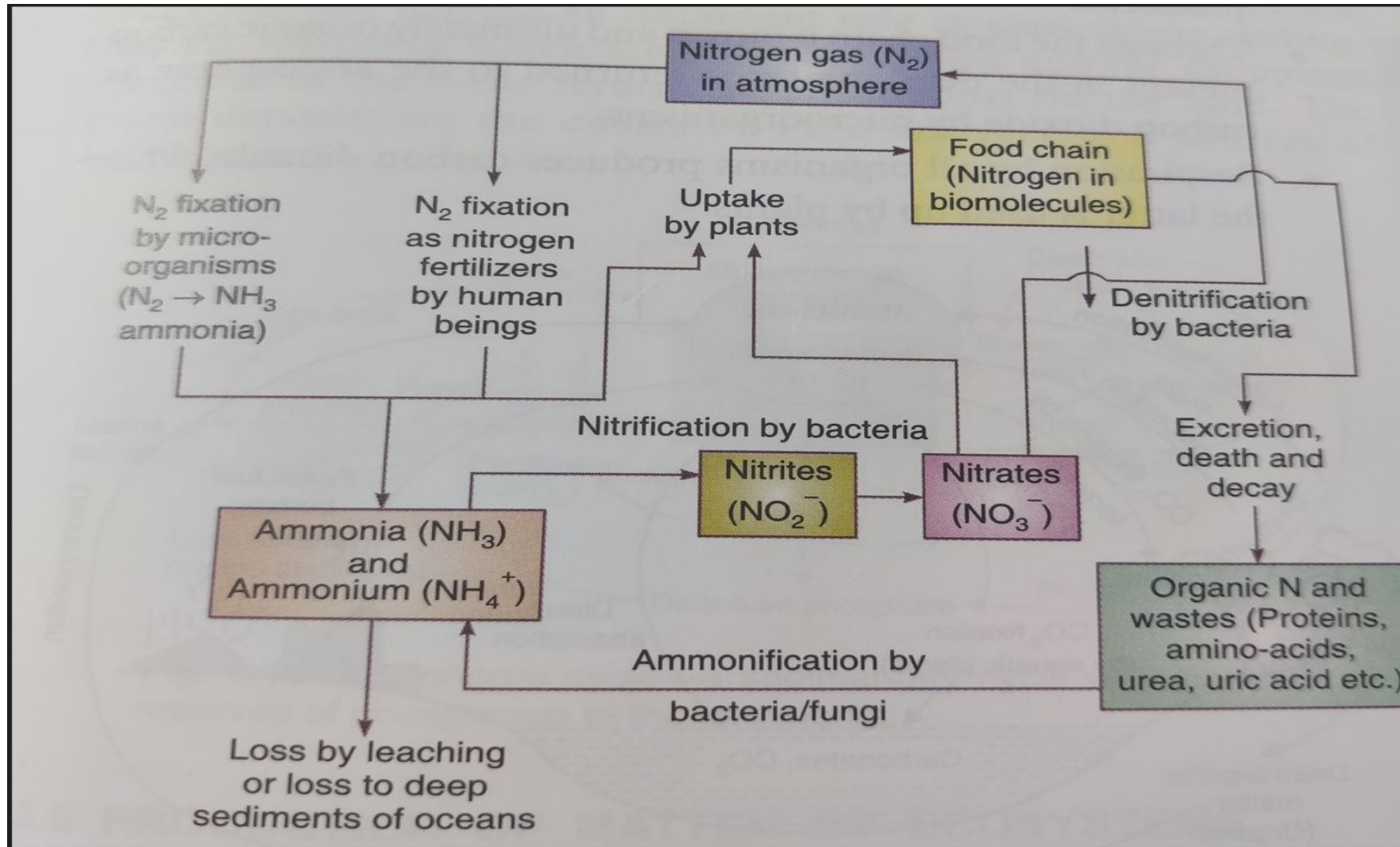
Nitrogen Fixation (NF) can be carried out by **symbiotic N fixer** and **non-symbiotic N fixer** and other natural as well as industrial process.



- **Symbiotic bacteria**, associated with roots of legumes and flowering plants. eg. **Rhizobium** which convert N into the organic nitrogen for their own cells. when these organisms die or leave wastes certain other bacteria and fungi return the N to the soils and atmosphere.
- **Non-symbiotic N fixers** are both aerobic and anaerobic bacteria as well as cyanobacteria. These occur in soil, marine and fresh water.
- **Lightning storms convert atmospheric N into nitrates and reaches the soil through rain water. They can also be converted to ammonia by denitrifying bacteria.**



Nitrogen cycle



PHOSPHOROUS CYCLE

- The only cycle that does not have a gaseous state, but trace amounts of gaseous phosphine {PH₃} do participate as a minor atmospheric component in the global biogeochemical phosphorus cycle.
- Inorganic phosphate PO_4^{3-} is released from rocks and sediments through the action of erosion. Soil PO_4^{3-} is absorbed by plants and incorporated into nucleic acids, phospholipids and ATP (adenosine triphosphate). Animals obtain most of their P by consumption of other animals and from water. PO_4^{3-} is released to the soil again by decomposers. Dissolved PO_4^{3-} gets absorbed by algae and aquatic plants. Decomposers break down waste and returns PO_4^{3-} to sediments on the seabed. Dead bodies and Ash immersion

**Soluble PO_4^{3-} in soil.
(orthophosphates)**

Bacteria

Parental Rock

Insoluble phosphate

**Plants
(Roots)/Algae**

Animals

Death

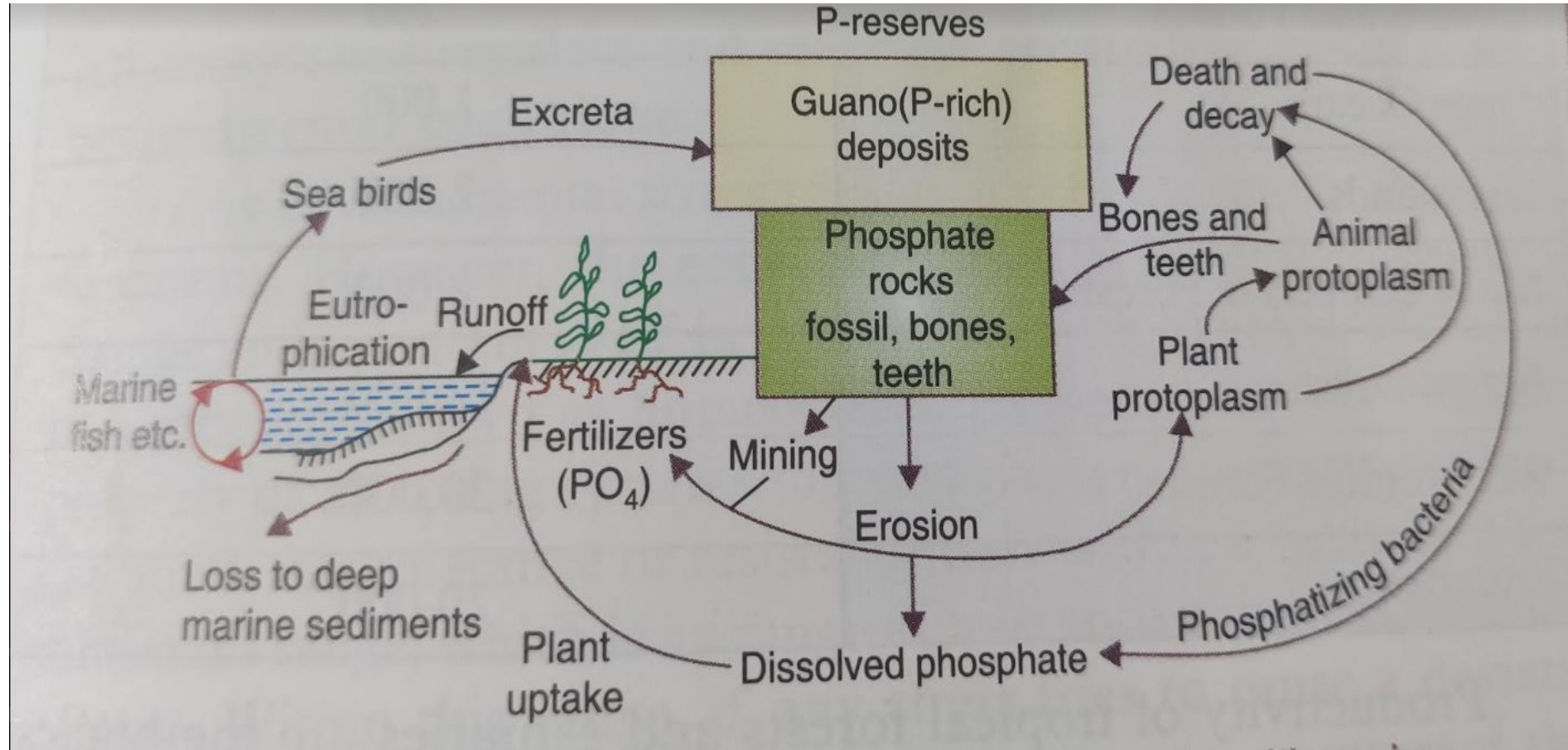
**Bacterial
Decomposition**

Bone/ Teeth

Decomposition

Phosphorus Cycle

Phosphorous cycle



Oxygen and Sulfur Cycle

Oxygen is the most import. element in our life.

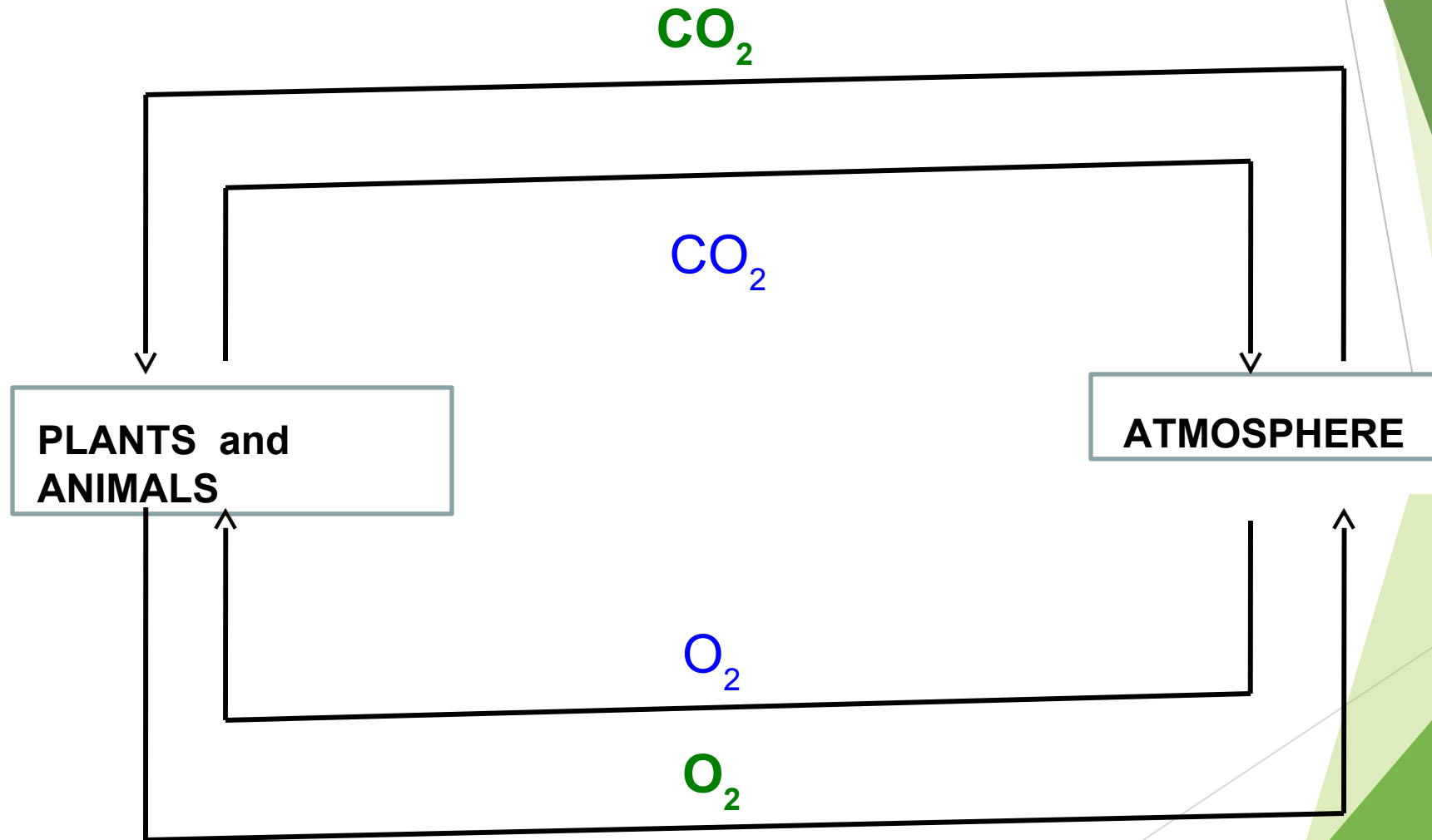
About 21% of Oxygen is present in the atm.

As free O_2 .

Plants and animals can take the free O_2 from the atmosphere through a process called respiration and it release CO_2 and water into the atm. Some plants do especially at night.

The sulphur cycle is a sedimentary biogeochemical cycle where sulphur moves through the earth, air, water, and living things. It transitions between mineral rocks, soil sulphates, atmospheric gases like sulphur dioxide, and organic proteins in living cells.

Oxygen Cycle



Video on Biogeochemical cycles

- ▶ https://www.google.com/search?q=video+on+biogeochemical+cycles&sca_esv=e0cc8f089a04fab1&rlz=1C1YTUH_enIN1045IN1045&sxsrf=APpeQnseuaTUHj4yEcCb2gnmcl_WTDsqDw:1786779622419&ei=5heAaqyhGbSqhvcP5d3ewQQ&start=10&sa=N&sstk=AU9db-Bf9YtQyCYYYKa9CiIMd8ml_9DpMZ4ZFnFxHrDuWSXvleO1-0LEvCGhRErzJhP-FWqJWegQsi8b981XI5dJqbWPenM_WjlEbA&ved=2ahUKEwisrL7lkkKWaxU0leEIHeWuN0gQ8tMDegQILRAE&biw=1507&bih=655&dpr=1.28#fpstate=ive&vld=cid:1fabb2b2,vid:C_qbmUAW-5c,st:0

Types of ecosystems

- ▶ On the basis of the presence of abiotic and biotic **factors** and their interaction amongst each other **there are 2 main ecosystems i.e., terrestrial or land and aquatic ecosystem.**
- ▶ As habitat varies on the earth with climate these are further subdivided into many subtypes.

Terrestrial ecosystem

- ▶ **Terrestrial ecosystem is basically the land ecosystem.** Major reservoirs are lithosphere and atmosphere.
- ▶ **Major terrestrial system is also called Biome**
- ▶ On the basis of the presence of biotic and abiotic factors the **terrestrial ecosystems are-**
- ▶ **the Forest ecosystem, Grassland ecosystem and Desert ecosystem.**

Forest ecosystem

- ▶ **Forest occupies approximately 40 % of total land on earth. In India forest cover is almost 21 % of its land area.**
- ▶ **These are the ecosystems having a predominance of trees that are interspersed with a large number of species of herbs, shrubs, climbers, lichens, algae and a wide variety of wild animals and birds.**



Producers: Different tree species



Consumers in a Forest Ecosystem



Decomposers in a Forest ecosystem

Components of forest ecosystem are as follows :

Abiotic components

- ▶ In addition to inorganic substances these also contain organic debris, e.g., fallen leaves, seeds, branches etc.
- ▶ It has canopy and sub-canopy , middle flora zone and ground flora zone, litter zone and rhizosphere zone with sun light penetration.

Biotic components

Living organisms present in the food chains.

Autotrophs (Producers)

Produce their food through photosynthesis. Trees are predominant species. There are herbs, shrubs, climbers, epiphytes e.g., orchids and ground flora.

There are 3 main forest types -

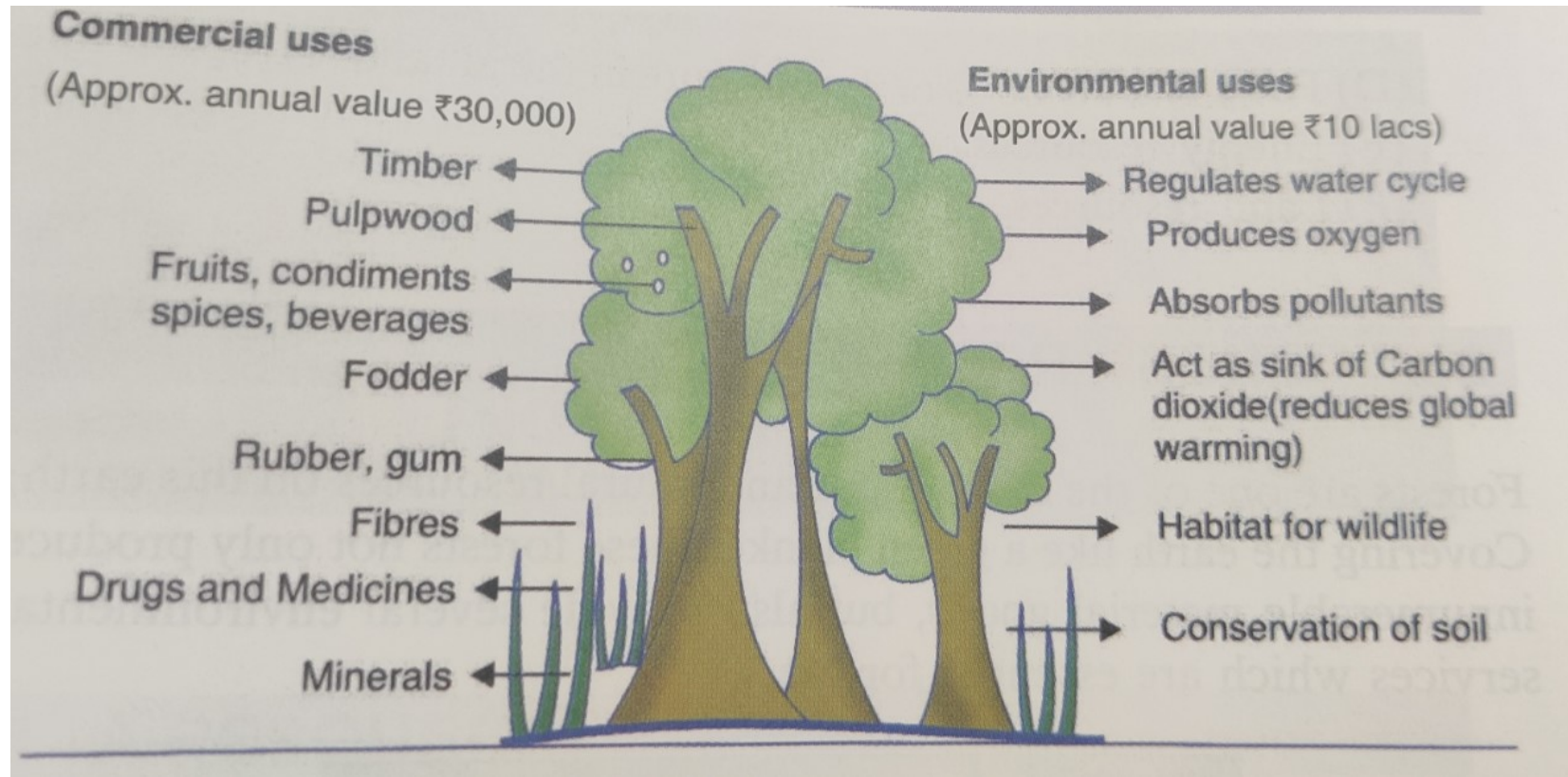
(1). Boreal (snow) forests, found farthest north, experience long, cold winters with short growing seasons.

(2). Temperate forests, located in the mid-latitudes mostly, have four distinct seasons. Mostly Coniferous vegetation/ no leaf shedding.

(3). Tropical (warm) forests, found along the equator, experience high temperatures and long growing seasons and harbor incredible amounts of biodiversity and are thicker than temperate forests, shed leaves annually.

Soft wood (pine) and Hard wood.

Economic vs Environmental value of a tree



Animals

- ▶ **Heterotrophs**

Herbivores, carnivores, omnivores (Consumer)

- a. **Primary consumers**

- ▶ These feed on the tree leaves as ants, flies, grasshoppers etc. Other larger animals feed on tree shoots like deer, Asian antelope (Nilgai), mongoose, bats etc.

- b. **Secondary consumers**

- ▶ Organisms depending on herbivores for their food e.g., fox, snakes, birds etc.

c. Tertiary consumers

- ▶ These are top carnivores depending on secondary consumers e.g., lion, tiger, man etc.

Saprotrophs (Decomposers)

- ▶ Various microorganisms such as fungi like *Aspergillus* spp., *Trichoderma* spp. etc. and bacteria such as *Bacillus* spp., *Clostridium* spp., *Pseudomonas* spp. etc. *Achromobacter*, *Acinetobacter*, *Alkanindiges*, *Alteromonas*, *Arthrobacter*, *Burkholderia*, *Dietzia*, *Enterobacter*, *Kocuria*, *Marinobacter* can degrade aromatic and aliphatic compounds.

- ▶ Thus these degrade organic chemicals for being recycled to nature through biological and biogeochemical processes.

GRASSLAND ECOSYSTEMS

- ▶ Grasslands are dominated by grass species but sometimes also allow the growth of a few trees and shrubs. Rainfall is average but erratic.
- ▶ Approximately 20-40 % of earth surface is grassland ecosystem.

Abiotic components

- ▶ Main abiotic factors are nutrients from earth and environment.

Biotic components

Autotrophs ; There are about a dozen different grasses,

- ▶ All grass forms such as Cyanadon spp/ Bermuda grass ., Pos spp./ flowering grass, Brachiaria spp/ animal feed., Dactyloctenium spp./ crowfoot grasses etc. along with herbs and shrubs. Switch grass, Miscanthus, Niper grass ?
- ▶ **Heterotrophs** : Herbivores and carnivores (consumers)

- ▶ These make available chemicals in nature and help in nutrient cycling.

Desert Ecosystem

- ▶ These type of ecosystem occupies 20-33 % land on earth.
- ▶ India- 12 %. Extreme weather conditions, less than 25 cm rainfall, high temperature, harsh climate are factors leading to desertification.
- ▶ **Abiotic components** : Sandy

- ▶ **Biotic components** : Desert biomass and animals.
- ▶ Less trees but mainly cactus type plants, shrubs, herbs, bushes, are dominant. Some species of lichens and mosses are also present.
- ▶ Plants in this region are known **as xerophytes** (cacti) etc
- ▶ **Heterotrophs (Consumers)**
- ▶ Herbivores and carnivores , the reptiles (snakes, lizards etc.), insects, birds and rodents (mice, squirrel etc.) which can sustain extreme weather conditions. Camel is the animal which is predominantly present. **Camel hump - fat storage.**

Biodegradations

- ▶ **Decomposers** : Some fungi and bacteria which can survive and survive on the plant species etc.

Aquatic ecosystem

- ▶ Interaction of biotic and abiotic components in hydrosphere.
- ▶ There are 3 main Aquatic ecosystems
 - ▶ 1. Freshwater ecosystem
 - ▶ 2. Marine ecosystem
 - ▶ 3. Astuary ecosystem

- ▶ Fresh water ecosystem is of 2 types:
 - ▶ Lotic ecosystem and lentic ecosystem,
 - ▶ Lotic / moving ecosystem - Flowing water- River, springs, streams etc.
-
- ▶ **Lentic /Still ecosystem (Pond ecosystem)**
 - ▶ Stagnant water
 - ▶ Small but several biotic and abiotic components are interrelated in several ways here.

Shorter Slide ?

- ▶ Abiotic components such as temperature, pH, light etc., organic and inorganic compounds such as amino acids, proteins, CO₂, O₂,
- ▶ Phosphate, nitrogen etc.

- ▶ **Biotic components**
- ▶ **Producers** : Photosynthetic bacteria (*Rhodospirillum rubrum*, *Cosmarium*, *Autotrophs spp.*, *Clostridium spp.*, *Spirulina* etc., phytoplankton and green plant. With the help of inorganic compounds these produce proteins, fats, carbohydrates etc.
- ▶ **Some plants are submerged in the ponds such as water hyacinth, Hydrilla etc.**

Fish, lizards, snakes - Ants- numbers - Collaborative Efforts - Traction to the niches

- ▶ **Consumers** : Some detritivores also eat dead plant matter. Mollusc, fish, frog etc. Fish eat algae and large fish eat small fish, frogs etc.
- ▶ **Decomposers** : Fungi and bacteria.
- ▶ **Lotic ecosystem (Rivers)** : Flowing water
 - ▶ **a.Producers** : Green algae, diatoms, mosses etc.
 - ▶ **b. Consumers** : Snails, flatworms, fish etc.
 - ▶ **c. Decomposers** : Fungi and bacteria.

Marine ecosystem

Marine ecosystem

- ▶ **Major ecosystem on earth as almost 70 % of earth surface is covered with water.**
- ▶ **Abiotic components** : Physicochemical factors are currents, waves, tides, light, temperature, pressure, salinity, etc.
- ▶ **Biotic components** :
- ▶ **Producers** :
- ▶ Microalgae, diatoms, diatoms, microalgae., macroalgae..

- ▶ **Consumers** : Herbivores such as fish larvae, copepods, molluscs, and carnivores such as Bombay duck, large fish, cod fish etc.
- ▶ **Decomposers** : Fungi and bacteria.

Estuary ecosystem

- ▶ Here river meets sea.

.

Allelopathy- Neem inhibits pests.

Eucalyptus- inhibits the growth

- ▶ **Abiotic components** : Change in salinity. Organisms can survive the change in salinity and there are more nutrients here
- ▶ **Biotic components**
- ▶ **Producers** : Sea weeds, Azolla, sea grass, algae etc. and mangroove/ mangrove forests (the trees and shrubs).
- ▶ Mangrooves can tolerate salinity.

- ▶ **Consumers** : Small fish, zooplanktons, large fish etc. Crustaceans and amphibians are also present.
- ▶ **Decomposers** : Fungi and bacteria.

PHOTOSYNTHESIS REACTION

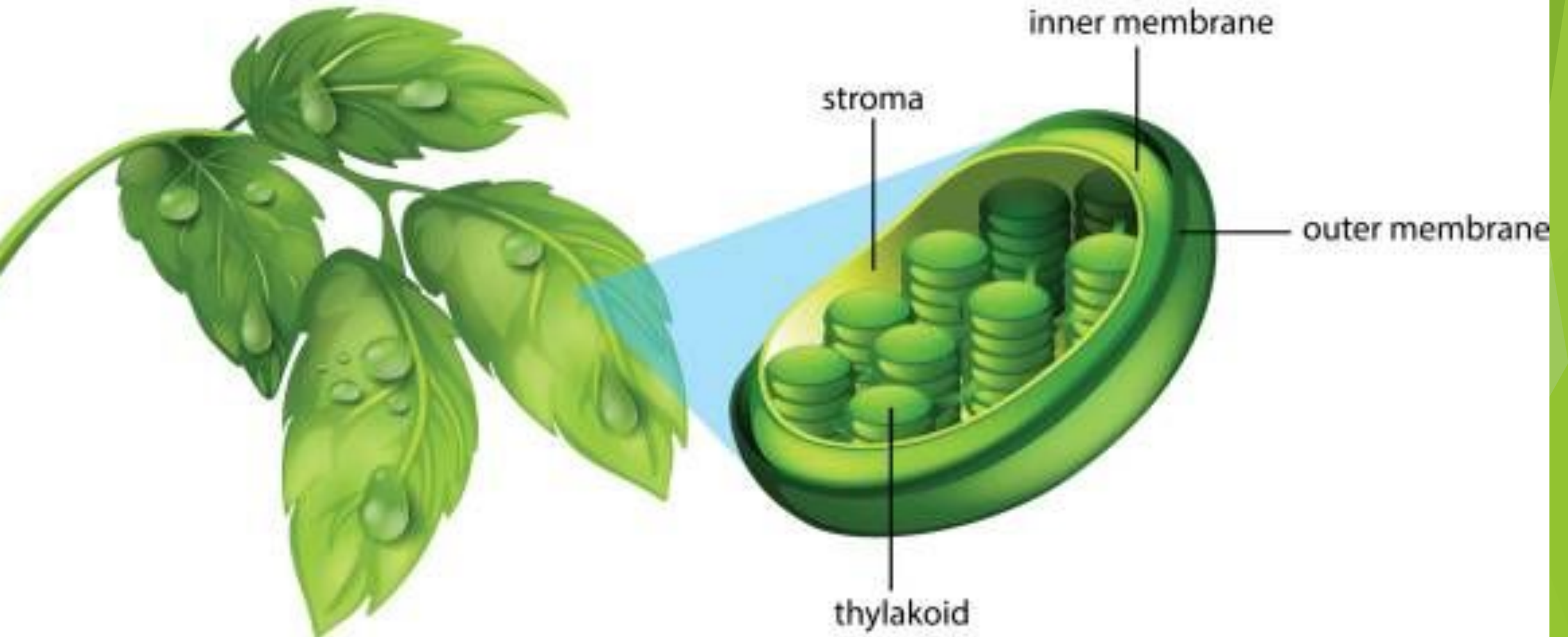
- ▶ Carbon dioxide is converted into sugars in a process called carbon fixation; photosynthesis captures energy from sunlight to convert carbon dioxide into carbohydrates. Carbon fixation is an endothermic redox reaction.
- ▶ Photosynthesis takes place in two distinct stages.

In the light reactions, energy from sunlight drives the synthesis of Adenosine triphosphate (ATP) and nicotinamide adenine dinucleotide phosphate (NADPH), coupled to the formation of O_2 from H_2O .
- ▶ In the dark reactions, so named because they do not require sunlight, the ATP and NADPH produced by the light reactions drive glucose synthesis.

PHOTOCHEMICAL PATHWAYS IN ECOSYSTEM

- ▶ **Photosynthesis**
- ▶ **Light dependent reactions**
- ▶ This reaction takes place in **chloroplast** of the cell.
- ▶ Light is converted into chemical energy in the form of ATP i.e., photophosphorylation and the photolysis of water takes place resulting in the formation of NADPH₂ and evolution of O₂..

Chloroplast in Plant Leaf



Light Independent Reactions

- ▶ This reaction takes place in the **stroma** of the chloroplast. By using products of **light dependent reaction** the CO₂ is fixed as glucose.
- ▶ Plants use some energy for their survival, activity and growth i.e., respiration. Remaining is used by herbivores for their activities. Then further the remaining is used by carnivores in the food chain.

- ▶ **OVERALL PHOTOSYNTHESIS REACTION**
- ▶ $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- ▶ Rubisco the Reversible Enzyme (Lithargic) is used.
- ▶ Carbonic anhydrase is an enzyme that facilitates the conversion between carbon dioxide (CO_2) and water (H_2O) to form carbonic acid (H_2CO_3) and its dissociative ions (HCO_3^- and H^+). It plays a key role in regulating pH and aiding in the transport of CO_2 in biological systems.

- **Sustainable development:** using resources to satisfy current needs without compromising future availability of resources
- Sustainability involves:
 - Renewable energy sources
 - Soil conservation, high-efficiency irrigation, organic agriculture
 - Pollution reduction
 - Habitat and species protection
 - Recycling
 - Fighting global climate change

- 
- The background of the slide features abstract, overlapping geometric shapes in various shades of green, ranging from light lime to dark forest green. These shapes are primarily located on the right side and bottom of the slide, creating a modern, dynamic feel.
- ▶ The United Nations sustainability framework centers on the 17 Sustainable Development Goals (SDGs), adopted by all UN member states in 2015 as part of the 2030 Agenda for Sustainable Development.

<https://www.un.org/sustainabledevelopment/sustainable-development-goals/>
Video



End of the slides/ Extra Relevant Information

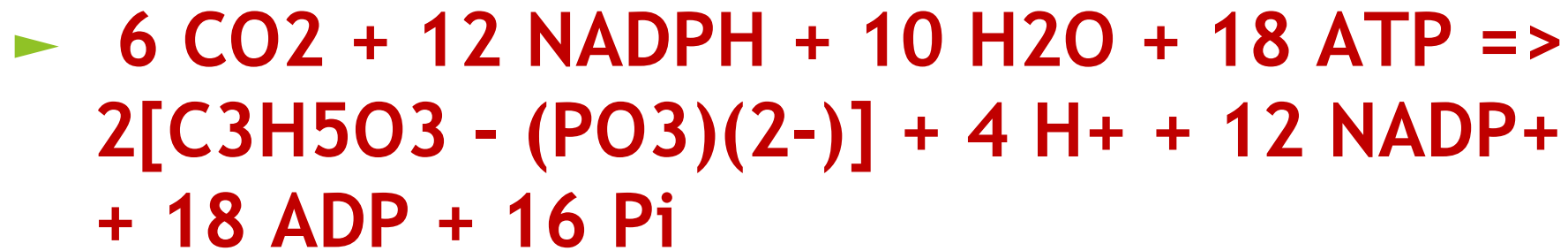
- ▶ **Respiration**
- ▶ **External respiration**
- ▶ Here oxygen is inhaled and carbon dioxide is exhaled. Gills and lungs are used in this gaseous exchange.
- ▶ **Internal Respiration**
- ▶ This is also called cellular respiration. Chemical energy in the organic molecules is released through oxidation in the form of ATP. When oxygen is required it is called aerobic cell respiration and when this takes place in the absence of O_2 this is called anaerobic cellular respiration.

- ▶ Organic molecules such as carbohydrates, proteins, fats , glycerides etc. are transferred to energy in the form of ATP, some is used and some is lost as heat. Such respiration takes place in almost every cell in the body.

Photorespiration

- ▶ The reaction takes place in the chlorophyll of the plant.
- ▶ Chlorophyll-b absorbs light and passes it on to chlorophyll-a where light energy is converted into chemical energy.

Biochemistry of Photosynthesis Reaction



Reactions in Bacteria

- ▶ $6 \text{ CO}_2 + 6 \text{ H}_2\text{S} = \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ S}$ (bacterial anaerobic photosynthesis)
- ▶ Actually, the balanced equation is: $6 \text{ CO}_2 + 12 \text{ H}_2\text{S} = \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ H}_2\text{O} + 12 \text{ S}$

- ▶ $6 \text{ CO}_2 + 18 \text{ ATP} + 12 \text{ NADPH} + 12 \text{ H}^+ \Rightarrow 2[\text{C}_3\text{H}_6\text{O}_3 - (\text{PO}_3\text{H})] + 18 \text{ ADP} + 16 \text{ Pi} + 12 \text{ NADP}^+ + 6 \text{ H}_2\text{O}$

- ▶ where

- ▶ $\text{Pi} = \text{PO}_3\text{H}$ (phosphate group)

- ▶ $\text{ATP} = \text{ADP} + \text{Pi}$

- ▶ $\text{NADPH} = \text{NADP}^+ + \text{H}^+ + 2\text{e}^-$

- ▶)

oxygenic photosynthesis

- ▶ The net-reaction of all light-dependent reactions in oxygenic photosynthesis is:
- ▶ $2\text{H}_2\text{O} + 2\text{NADP}^{++} + 3\text{ADP} + 3\text{P}_i \rightarrow \text{O}_2 + 2\text{H}^+ + 2\text{NADPH} + 3\text{ATP}$

Light independent reactions

- ▶ This reaction takes place in the stroma of the chloroplast. By using products of light dependent reaction the CO_2 is fixed as glucose.
- ▶ The products of the light independent reaction are glucose, NADP^+ , and H_2O . The reactants are CO_2 , NADPH , and ATP .
- ▶ Plants use some energy for their survival i.e., respiration. Remaining is used by herbivores for their activities. Remaining is used by carnivores in the food chain.

Respiration

- ▶ **External respiration**
- ▶ Here oxygen is inhaled and carbon dioxide is exhaled. Gills and lungs are used in this gaseous exchange.

Photorespiration

- ▶ The reaction takes place in the chlorophyll of the plant.
- ▶ Chlorophyll-b absorbs light and passes it on to chlorophyll-a where light energy is converted into chemical energy.

Internal Respiration

- ▶ This is also called cellular respiration. Chemical energy in the organic molecules is released through oxidation in the form of ATP. When oxygen is required it is called aerobic cell respiration and when this takes place in the absence of O_2 this is called anaerobic cellular respiration. Organic molecules such as carbohydrates, proteins, fats, glycerides etc. are transferred to energy in the form of ATP, some is used and some is lost as heat. Such respiration takes place in almost every cell in the body.

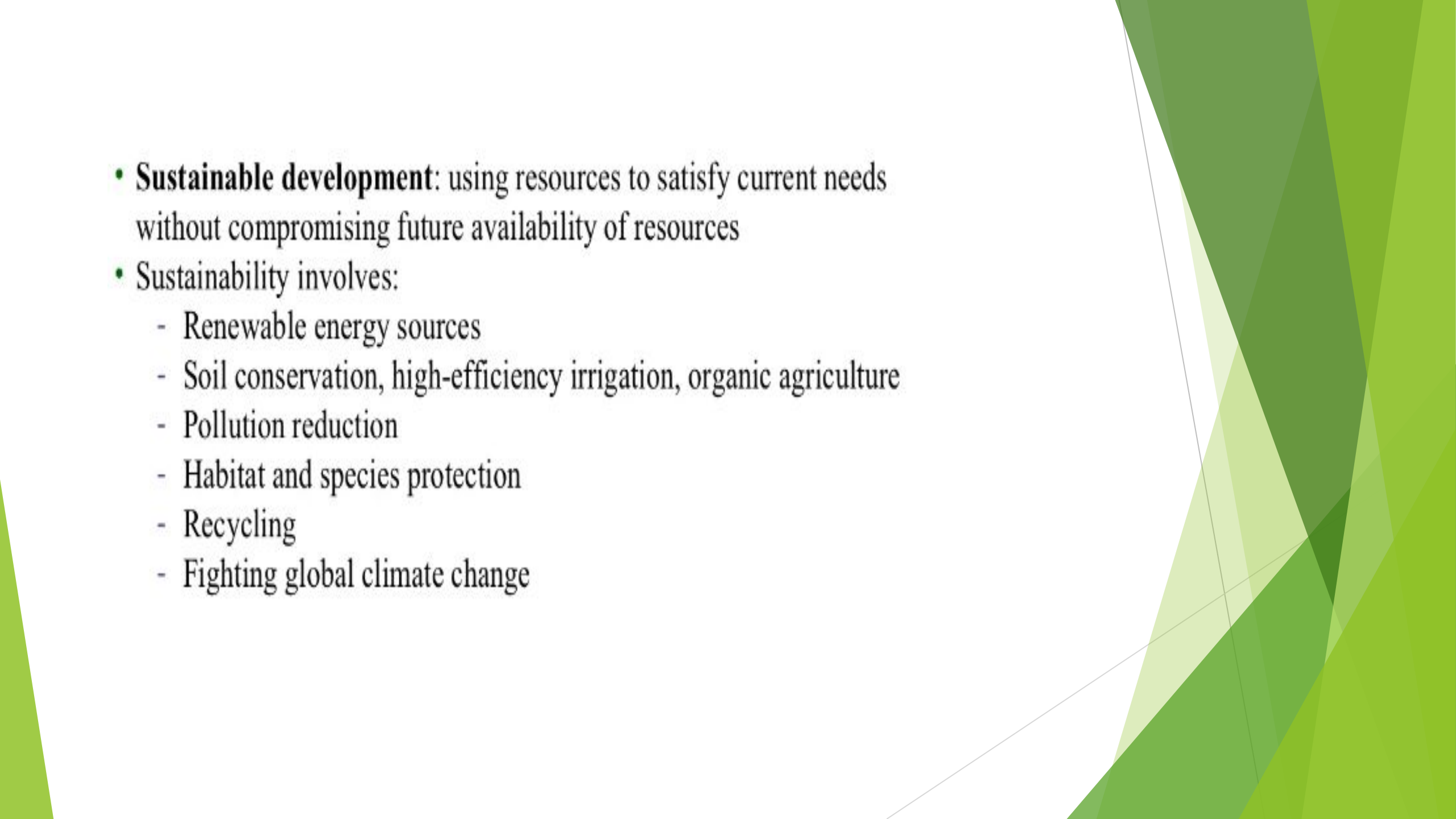
- 
- The background of the slide features abstract, overlapping geometric shapes in various shades of green, ranging from light lime to dark forest green. These shapes are primarily located on the right side of the slide, creating a modern, layered effect.
- **Sustainable development:** using resources to satisfy current needs without compromising future availability of resources
 - Sustainability involves:
 - Renewable energy sources
 - Soil conservation, high-efficiency irrigation, organic agriculture
 - Pollution reduction
 - Habitat and species protection
 - Recycling
 - Fighting global climate change

Table 4.1. India's major biogeographic habitats

145

Sr. No.	Biogeographic zone	Biotic province	Total area (Sq. Km.)	Important flora and fauna
1.	Trans-Himalayan	Upper Regions	186200	• Pine, Deodar • Wild sheep, Yak, Tibetan ass, Snow Leopard, Marmot, Wolf, Black necked crane.
2.	Himalayan	North-West Himalayas West Himalayas Central Himalayas East Himalayas	6900 720000 123000 83000	• Pine, Cork tree, Sal, Dhaak (Butea), Castor • Wild bear, Sambar, Leopard, Sikkim Stag, Musk Deer
3.	Desert	Kutch Thar Ladakh	45000 180000 NA	• Acacia, Zizyphus, Khejri, Date palm. • Camel, Bustard, Wild ass, Desert cat, Fox, Rat
4.	Semi-arid	Central India Gujarat- Rajwara	107600 400400	• Acacia, Date palm, Peepal • Gir Lion, Tiger Sariska and Ranthambore (Tiger Reserves)
5.	Western Ghats	Malabar Coast Western Ghat Mountains	59700 99300	• Sheesham, Peepal, Tuna, Bahera • Tortoise, Frog, Lizards, Snakes
6.	Deccan Peninsula	Deccan Plateau South-Central Plateau Eastern Plateau Chhota Nagpur Central Highlands	378000 341000 198000 217000 287000	• Acacia, Palaash, Tuna, Pine, Castor • Sambar, Sloth bear, Tiger, Cheetal, Four-horned stag, Wild elephant, Wild buffalo

(Contd.)

7.	Gangetic Plain	Upper Gangetic Plain	206400	• Sal, Acacia, Jamun, Mango, Bael
		Lower Gangetic Plain	153000	• Black chinkara, Stag, Rhinoceros, Gazzel, Alligator, Turtle.
8.	North-East India	Brahmaputra Valley	65200	• Bamboo, Sal, Jack fruit, Tuna Chestnut, Castor
		North-Eastern Hills	106200	• Elephant, Rhinoceros, Yak, Deer, Porcupine
9.	Islands	Andaman Islands	6397	• Bahera, Harar, Jack fruit, Cardamom, Coconut, Cloves
		Nicobar Islands	1930	• Dolphin, Alligator, Molluscs
		Lakshadweep Islands	180	
10.	Coasts	West Coast	6500	• Coconut, Banana, Cashew nut
		East Coast	6500	• Dugong, Dolphin, Turtle, Alligator, Molluscs

Table 4.5 Global Hot spots of biodiversity

Hot spots	Plant species	Endemic plants	% of Global plants	Vertebrate species	Endemic vertebrates	% of Global vertebrates
1. Tropical Andes	45000	20000	6.7	3389	1567	5.7
2. Mesoamerican Forests	24000	5000	1.7	2859	1159	4.2
3. Caribbean	12000	7000	2.3	1518	779	2.9
4. Brazil's Atlantic Forest	20000	8000	2.7	1361	567	2.1
5. Choc/Darien of Panama Western Ecuador	9000	2250	0.8	1625	418	1.5
6. Brazil's Cerrado	10000	4400	1.5	1268	117	0.4
7. Central Chile	3429	1605	0.5	335	61	0.2
8. California Floristic Province	4426	2125	0.7	584	71	0.3
9. Madagascar	12000	9704	3.2	987	771	2.8
10. Eastern Arc and Coastal Forest of Tanzania/Kenya	4000	1500	0.5	1019	121	0.4
Western African Forests	9000	2250	0.8	1320	270	1.0
Cape Floristic Province	8200	5682	1.9	562	53	0.2
Succulent Karoo	4849	1940	0.6	472	45	0.2
Mediterranean Basin	25000	13000	4.3	770	235	0.9

	Population Reduction Rates	Geographic Range		Population Size	Population Restrictions	Extinction Probability (in the wild)
		Extent of Occurrence	Area of Occupancy			
Least Concern	A species that has a widespread and abundant population					
Near Threatened	A species that is likely to qualify for a threatened category in the near future					
Vulnerable Species	30-50% population decline	<20,000 km ²	<2,000 km ²	<10,000 mature individuals	<1,000 mature individuals or an area of occupancy of <20km ²	at least 10% within 100 years
Endangered Species	50-70% population decline	<5,000 km ²	<500 km ²	<2,500 mature individuals	<250 mature individuals	at least 20% within 20 years or 5 generations
Critically Endangered	>80-90% population decline	<100 km ²	<10 km ²	<250 mature individuals	<10 mature individuals	at least 50% within 10 years or 3 generations
Extinct in the wild	Only survives in cultivation (plants), in captivity (animals), or as a population well outside its established range					
Extinct	No remaining individuals of the species					